# Functional Genomics 11: RNAseq

Michael Schatz

Oct 26 2026

Lecture 16. Applied Comparative Genomics



# Project Proposal
# Due Wednesday Oct 21 @ 11:59pm

## Project Proposal

Assignment Date: Monday October 12, 2026
Due Date: Wednesday, October 21 2026 @ 11:59pm

Review the Project Ideas page

Work solo or form a team for your class project of no more than 3 people.

The proposal should have the following components:

- Name of your team
- List of team members and email addresses
- Short title for your proposal
- 1 to 2 paragraph description of what you hope to do and how you will do it
- References to 3 to 4 relevant papers
- References/URLs to datasets that you will be studying (Note you can also use simulated data)
- Please add a note if you need me to sponsor you for a MARCC/skipjack account (high RAM, GPUs, many cores, etc)
- Please add a note if you need me to sponsor you for an AnVIL cloud billing account (dynamic resources)

Submit the proposal as a 1 to 2 page PDF on GradeScope (each team should submit one proposal and tag all people in the team). After submitting your proposal, I will provide feedback. If necessary, we can schedule a time to discuss your proposal, especially to ensure you have access to the data that you need. The sooner that you submit your proposal, the sooner we can schedule the meeting.

Later, you will present your project in class during the last week of class. You will also submit a written report (5-7 pages) of your project, formatting as a Bioinformatics article (Intro, Methods, Results, Discussion, References).

Please use Piazza to coordinate proposal plans!

# Assignment 4: RNA-seq and Machine Learning

Assignment Date: Monday, October 26, 2026
Due Date: Monday, November 2 @ 11:59pm

## Assignment Overview

In this assignment you will explore a couple of key aspects of RNA-seq and introduce the key concepts of machine learning. For this assignment, we will provide a Jupyter notebook with code for you to use and complete your assignment in.

To use the Jupyter notebook, create a new mamba environment for assignment 4:

```
mamba create -n asn4 scikit-learn pytorch matplotlib jupyter umap-learn seaborn pandas numpy
```

Make sure to download the training data for the assignment from Dropbox. Place this in the `data_files` directory and make sure to uncompress ( `gunzip` ) both the training and testing files.

As a reminder, any questions about the assignment should be posted to Piazza

See the notebook here: Assignment4.ipynb

## Packaging

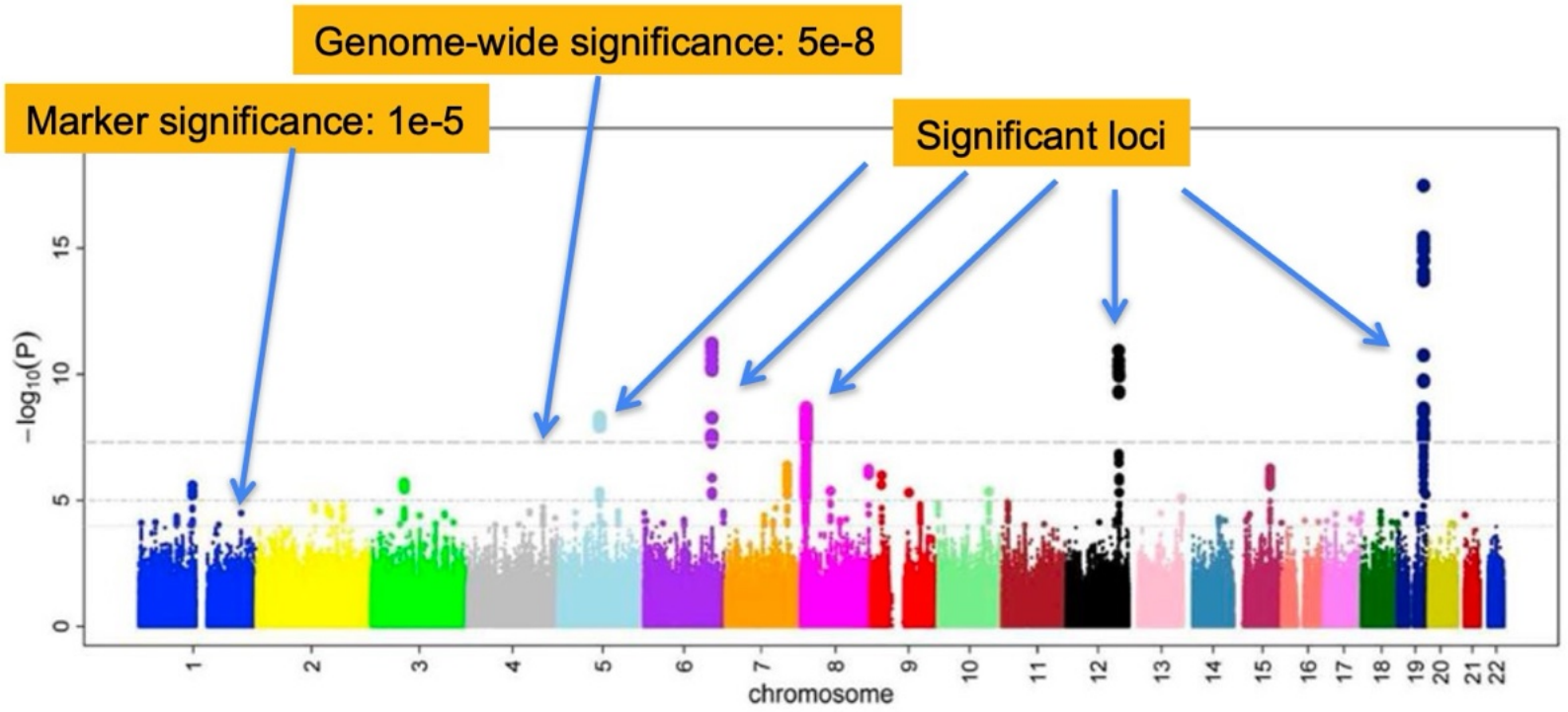
The solutions to the above questions should be submitted as a single PDF document that includes your name, email address, and all relevant figures (as needed). If you use ChatGPT for any of the code, also record the prompts used. Submit your solutions by uploading the PDF to GradeScope, and remember to select where in your submission each question/subquestion is.

If you submit after this time, you will use your late days. Remember, you are only allowed 4 late days for the entire semester!

## Resources

- Jupyter notebooks: https://jupyter.org/
- scikit-learn: https://scikit-learn.org/stable/
- pytorch: https://pytorch.org/
- umap-learn: https://umap-learn.readthedocs.io/en/latest/basic_usage.html

# Manhattan Plot




- The conventional genome-wide significance threshold is p-value ≤ $5 \times 10^{-8}$
- This threshold corresponds to a Bonferroni correction for approximately one million independent SNP tests ($0.05 / 10^6$).
- Nearby SNPs often form association peaks due to linkage disequilibrium (LD), as correlated variants tend to be inherited together.

Ikram et al, PLOS Genetics. 2010

# GWAS Catalog

As of 2026-10-03, the GWAS Catalog contains 7845 publications,
1,198,394 top associations and 203,002 full summary statistics.

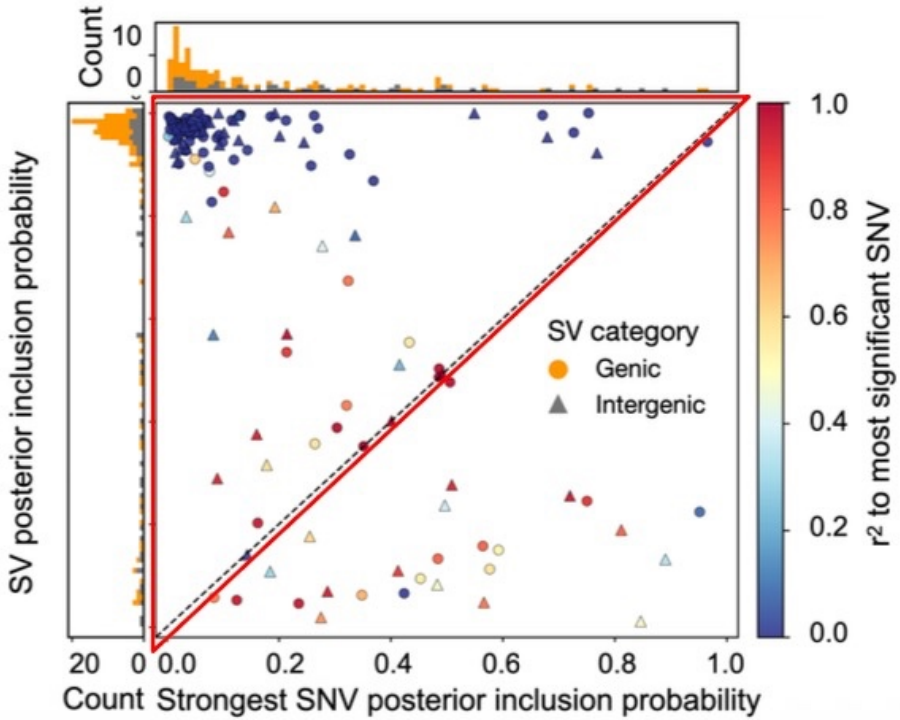


- Abdominal aortic aneurysm
- Acute lymphoblastic leukemia
- Adhesion molecules
- Adverse response to carbamapezine
- Adiponectin levels
- Age-related macular degeneration
- AIDS progression
- Alcohol dependence
- Alopecia areata
- Alzheimer disease
- Amyloid A levels
- Amyotrophic lateral sclerosis
- Angiotensin-converting enzyme activity
- Ankylosing spondylitis
- Arterial stiffness
- Asparagus anosmia
- Asthma
- Atherosclerosis in HIV
- Atrial fibrillation
- Attention deficit hyperactivity disorder
- Autism
- Basal cell cancer
- Behcet's disease
- Bipolar disorder
- Biliary atresia
- Bilirubin
- Bitter taste response
- Birth weight
- Bladder cancer
- Bleomycin sensitivity
- Blond or brown hair
- Blood pressure
- Blue or green eyes
- BMI, waist circumference
- Cleft lip/palate
- Cognitive function
- Conduct disorder
- Colorectal cancer
- Corneal thickness
- Coronary disease
- Creutzfeldt-Jakob disease
- Crohn's disease
- Cutaneous nevi
- Dermatitis
- Drug-induced liver injury
- Endometriosis
- Eosinophil count
- Eosinophilic esophagitis
- Erectile dysfunction and prostate cancer treatment
- Erythrocyte parameters
- Esophageal cancer
- Essential tremor
- Exfoliation glaucoma
- Eye color traits
- F cell distribution
- Fibrinogen levels
- Folate pathway vitamins
- Follicular lymphoma
- Fuch's corneal dystrophy
- Freckles and burning
- Gallstones
- Gastric cancer
- Glioma
- Glycemic traits
- Hair color
- Hair morphology
- Handedness in dyslexia
- HDL cholesterol

http://www.ebi.ac.uk/gwas/diagram

## Genome-Wide SV-Disease Associations

Comparison of the association significance between each SV and the strongest nearby SNV (within ±100 kb) for the same phenotype




After fine-mapping:

- SVs were included in the 80% casual set in 138 (92.0%) associations.
- 120 associations (80.0 %) where the SV was the most likely causal variant within the locus.

# Genetic Basis of Autism Spectrum Disorders



***Complex disorders of brain development***

- Characterized by difficulties in social interaction, verbal and nonverbal communication and repetitive behaviors.
- Have their roots in very early brain development, and the most obvious signs of autism and symptoms of autism tend to emerge between 2 and 3 years of age.

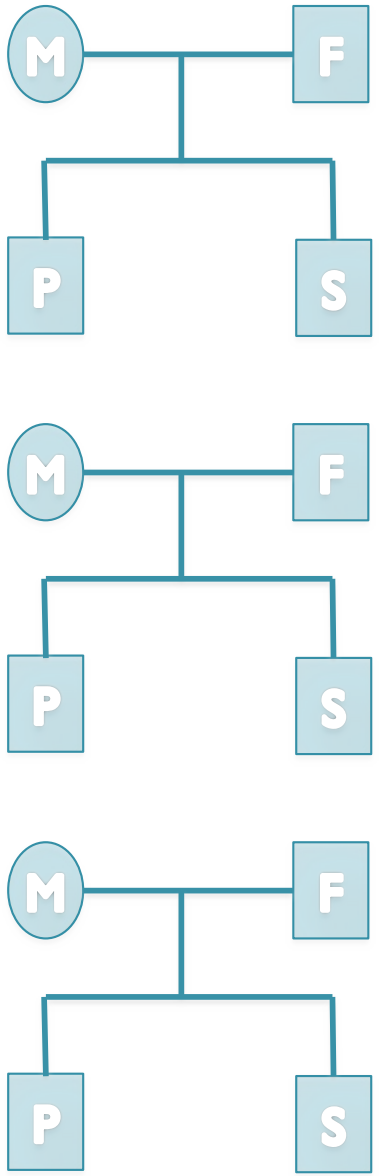
***U.S. CDC identify around 1 in 68 American children as on the autism spectrum***

- Ten-fold increase in prevalence in 40 years, only partly explained by improved diagnosis and awareness.
- Studies also show that autism is four to five times more common among boys than girls.
- Specific causes remain elusive

# Searching for the genetic risk factors

***Search Strategy***

- Thousands of families identified from a dozen hospitals around the United States
- Large scale genome sequencing of "simplex" families: mother, father, affected child, unaffected sibling
- Unaffected siblings provide a natural control for environmental factors

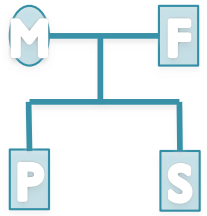



***Are there any genetic variants present in affected children, that are not in their parents or unaffected siblings?***

SFARI SIMONS FOUNDATION AUTISM RESEARCH INITIATIVE

# De novo mutation discovery and validation

**De novo mutations**:
Sequences not inherited from your parents.




```
Reference:    ...TCAAATCCTTTTAATAAAGAAGAGCTGACA...

Father(1):    ...TCAAATCCTTTTAATAAAGAAGAGCTGACA...
Father(2):    ...TCAAATCCTTTTAATAAAGAAGAGCTGACA...

Mother(1):    ...TCAAATCCTTTTAATAAAGAAGAGCTGACA...
Mother(2):    ...TCAAATCCTTTTAATAAAGAAGAGCTGACA...

Sibling(1):   ...TCAAATCCTTTTAATAAAGAAGAGCTGACA...
Sibling(2):   ...TCAAATCCTTTTAATAAAGAAGAGCTGACA...

Proband(1):   ...TCAAATCCTTTTAATAAAGAAGAGCTGACA...
Proband(2):   ...TCAAATCCTTTTAAT****AAGAGCTGACA...
```

4bp heterozygous deletion at chr15:93524061 CHD2

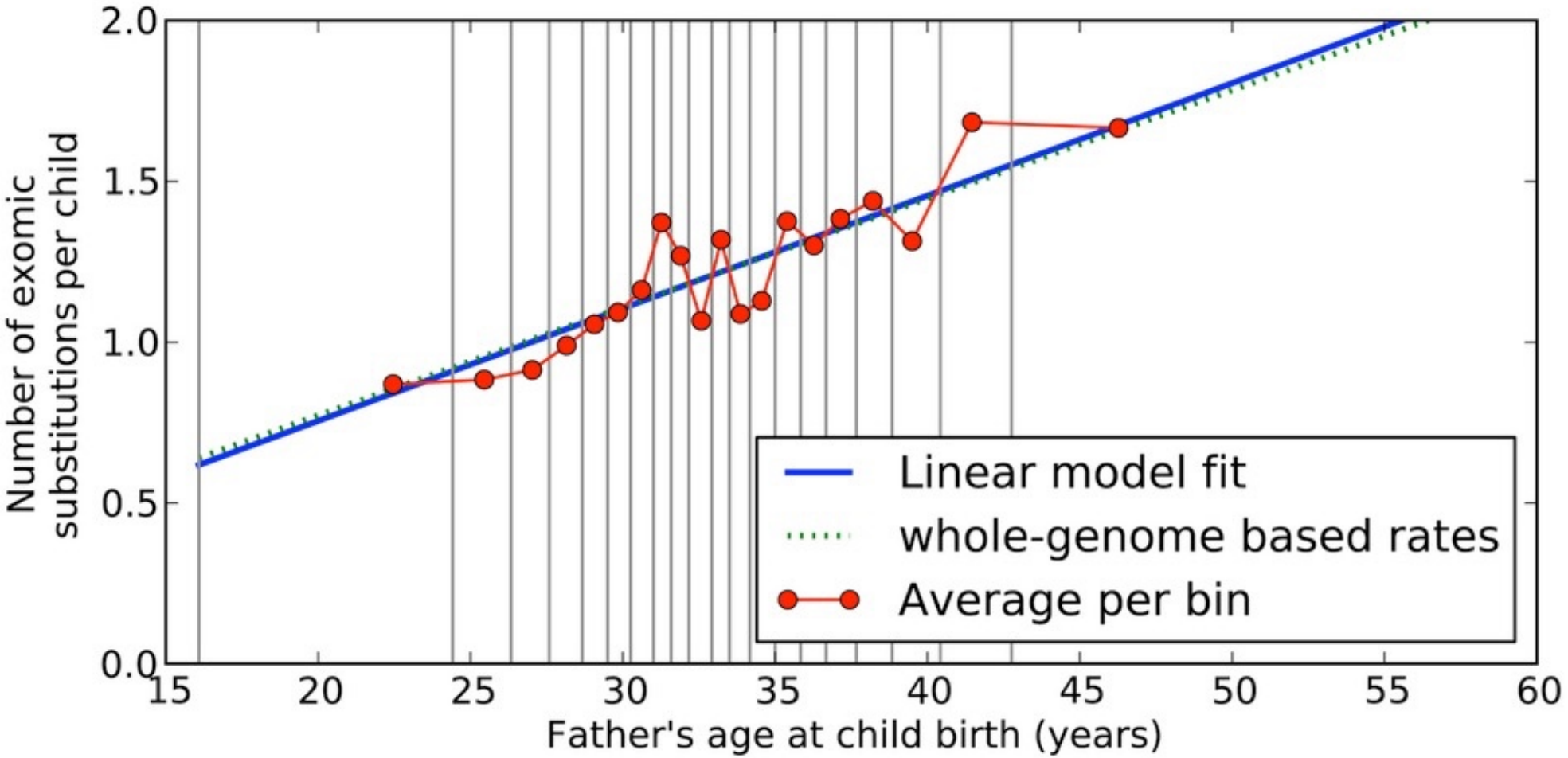
# De novo Genetics of Autism

- In 593 family quads so far, we see significant enrichment in de novo ***likely gene killers*** in the autistic kids
  - Overall rate basically 1:1
  - 2:1 enrichment in nonsense mutations
  - 2:1 enrichment in frameshift indels
  - 4:1 enrichment in splice-site mutations
  - Most de novo originate in the paternal line in an age-dependent manner (56:18 of the mutations that we could determine)

- Observe strong overlap with the 842 genes known to be associated with fragile X protein FMPR
  - Related to neuron development and synaptic plasticity
  - Also strong overlap with chromatin remodelers

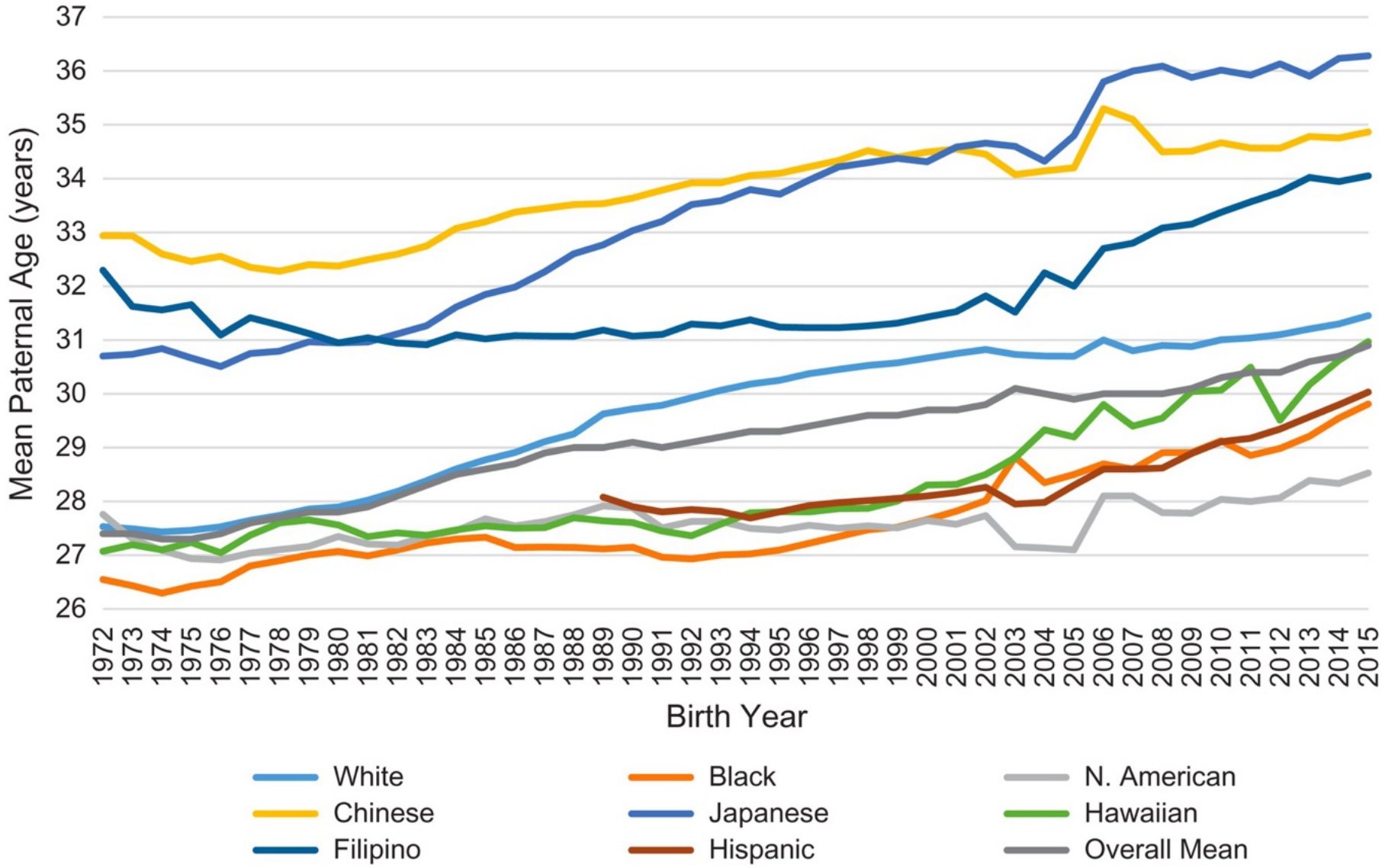
**Accurate de novo and transmitted indel detection in exome-capture data using microassembly.**
Narzisi *et al* (2014) Nature Methods doi:10.1038/nmeth.3069

# De novo Mutations in Men




**The contribution of de novo coding mutations to autism spectrum disorder**
Iossifov *et al* (2014) *Nature*. doi:10.1038/nature13908

# Age of Fatherhood




**The age of fathers in the USA is rising: an analysis of 168 867 480 births from 1972 to 2015**
Khandwala *et al* (2017) *Human Reproduction. https://doi.org/10.1093/humrep/dex267*

# Goal: Genome Annotations

aatgcatgcggctatgctaatgcatgcggctatgctaagctgggatccgatgacaatgcatgcggctatgctaat
gcatgcggctatgcaagctgggatccgatgactatgctaagctgggatccgatgacaatgcatgcggctatgct
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gctatgctaatgcatgcggctatgctaagctcatgcgg

# Goal: Genome Annotations

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gcatgcggctatgcaagctgggatccgatgactatgctaagctgggatccgatgacaatgcatgcggctatgct
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gctatgctaatgcatgcggctatgctaagctcatgcgg



# Outline

1. Alignment to other genomes
2. Prediction aka "Gene Finding"
3. Experimental & Functional Assays

# Seed and Extend

```
FAKDFLAGGVAAAISKTAVAPIERVKLLLQVQHASKQITADKQYKGIIDCVVRIPKEQGV
|   |     | |||  |||||||||||||||||| |||  |   |   ||||  |    | ||||||
FLIDLASGGTAAAVSKTAVAPIERVKLLLQVQDASKAIAVDKRYKGIMDVLIRVPKEQGV
```

- Homologous sequences are likely to contain a short high scoring word pair, a seed.
  - Smaller seed sizes make the sense more sensitive, but also (much) slower
  - Typically do a fast search for prototypes, but then most sensitive for final result

- BLAST then tries to extend high scoring word pairs to compute high scoring segment pairs (HSPs).
  - Significance of the alignment reported via an e-value

# BLAST E-values

E-value = the number of HSPs having alignment score S (or higher) expected to occur by chance.

→ Smaller E-value, more significant in statistics

→ Bigger E-value, less significant

→ Over 1 means expect this totally by chance

(not significant at all!)

The expected number of HSPs with the score at least S is :

$$E = K*n*m*e^{-\lambda S}$$

K, λ are constant depending on model

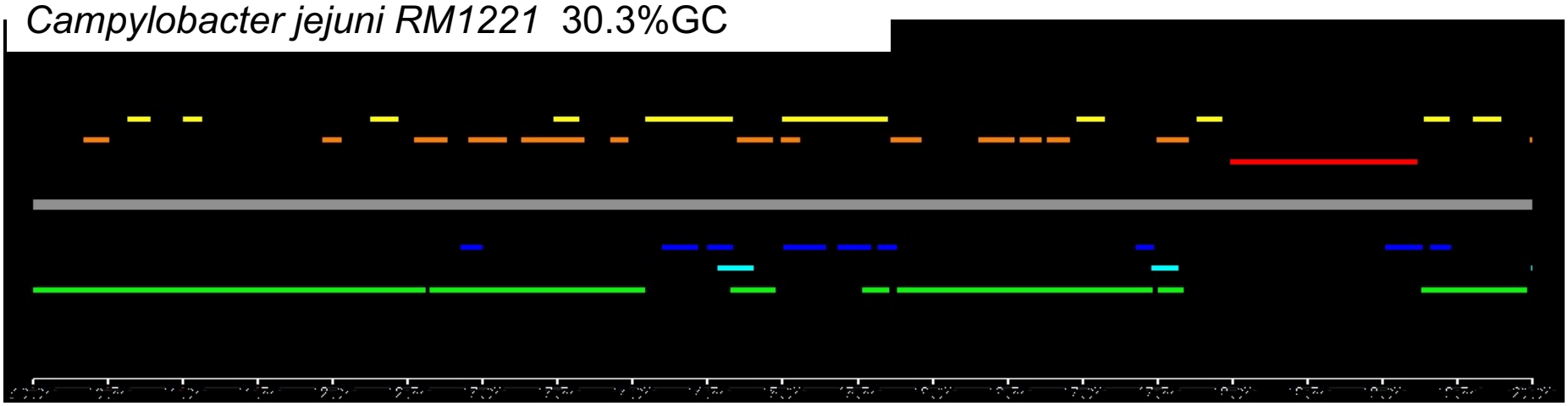
n, m are the length of query and sequence

E-values quickly drop off for better alignment bits scores



# Outline

1. Alignment to other genomes
2. Prediction aka "Gene Finding"
3. Experimental & Functional Assays



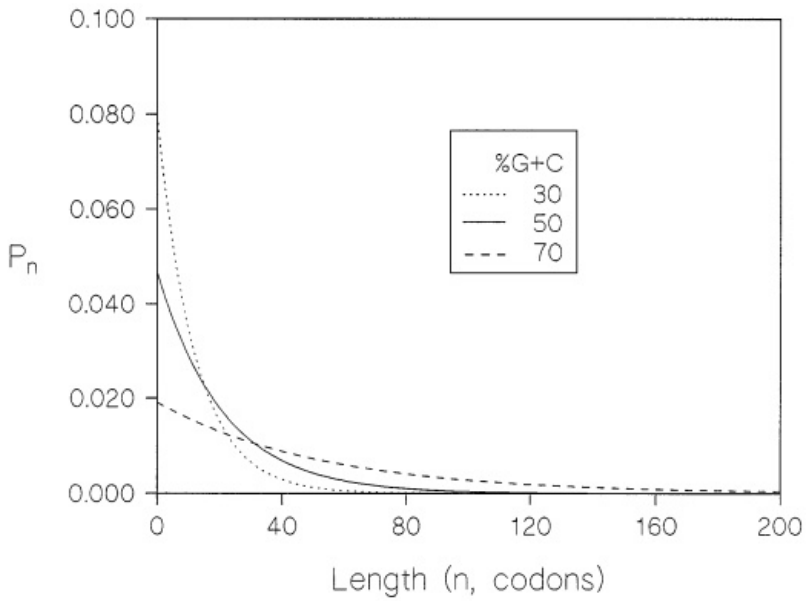

All ORFs longer than 100bp on both strands shown
- color indicates reading frame

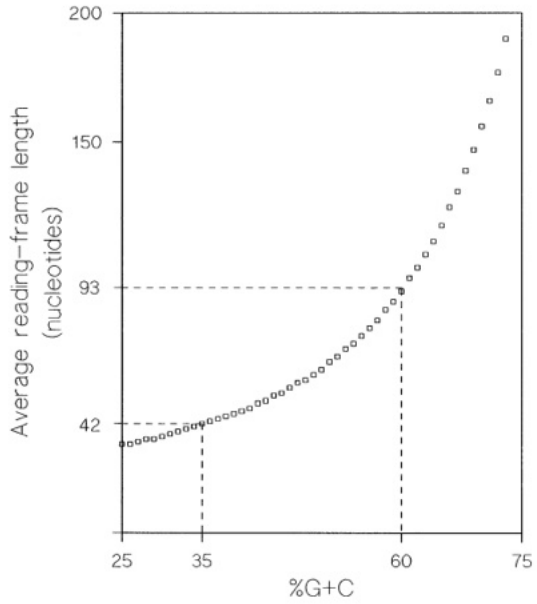
Longest ORFs likely to be protein-coding genes

Note the low GC content

All genes are ORFs but not all ORFs are genes

# Stop Codon Frequencies




***If the sequence is mostly A+T, then likely to form stop codons by chance!***

***In High A+T (Low G+C):***
Frequent stop codons; Short Random ORFs; long ORFs likely to be true genes

***In High G+C (Low A+T):***
Rare stop codons; Long Random ORFs; harder to identify true genes

***A relationship between GC content and coding-sequence length.***
Oliver & Marín (1996) J Mol Evol. 43(3):216-23.

# Probabilistic Methods

- Create models that have a probability of generating any given sequence.
  - Evaluate gene/non-genome models against a sequence
- Train the models using examples of the types of sequences to generate.
  - Use RNA sequencing, homology, or "obvious" genes
- The "score" of an orf is the probability of the model generating it.
  - Most basic technique is to count how kmers occur in known genes versus intergenic sequences
  - More sophisticated methods consider variable length contexts, "wobble" bases, other statistical clues



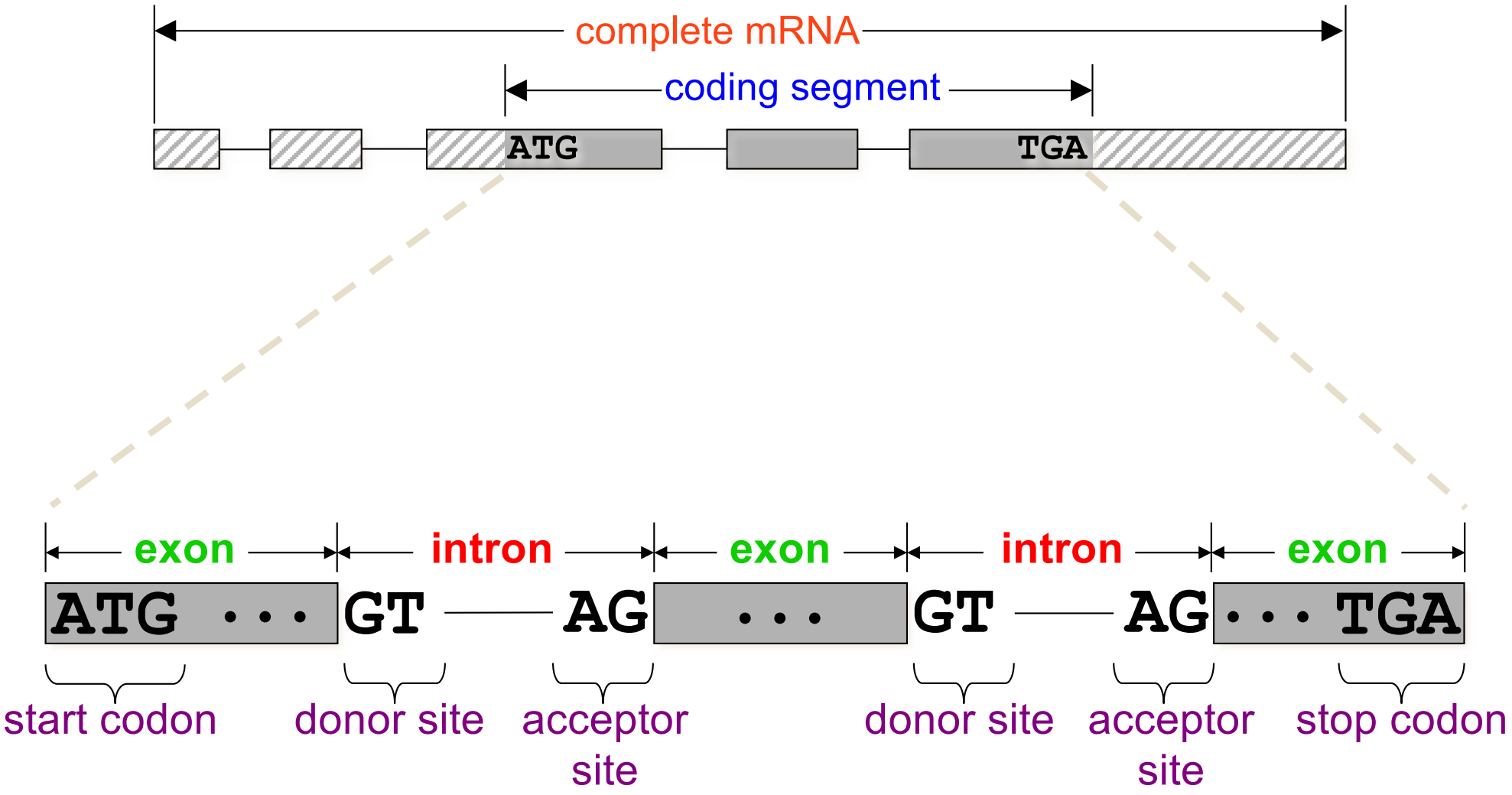
# Overview of Eukaryotic Gene Prediction

CBB 231 / COMPSCI 261

*W.H. Majoros*

Duke
UNIVERSITY

# Eukaryotic Gene Syntax

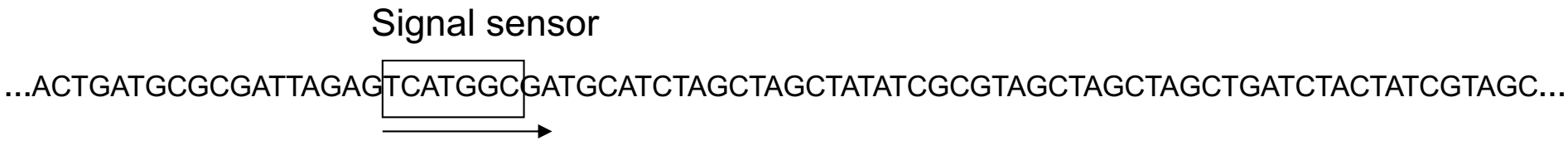



Regions of the gene outside of the CDS are called *UTR*'s (*untranslated regions*), and are mostly ignored by gene finders, though they are important for regulatory functions.

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# Identifying Signals In DNA

We slide a fixed-length model or "window" along the DNA and evaluate `score(signal)` at each point:

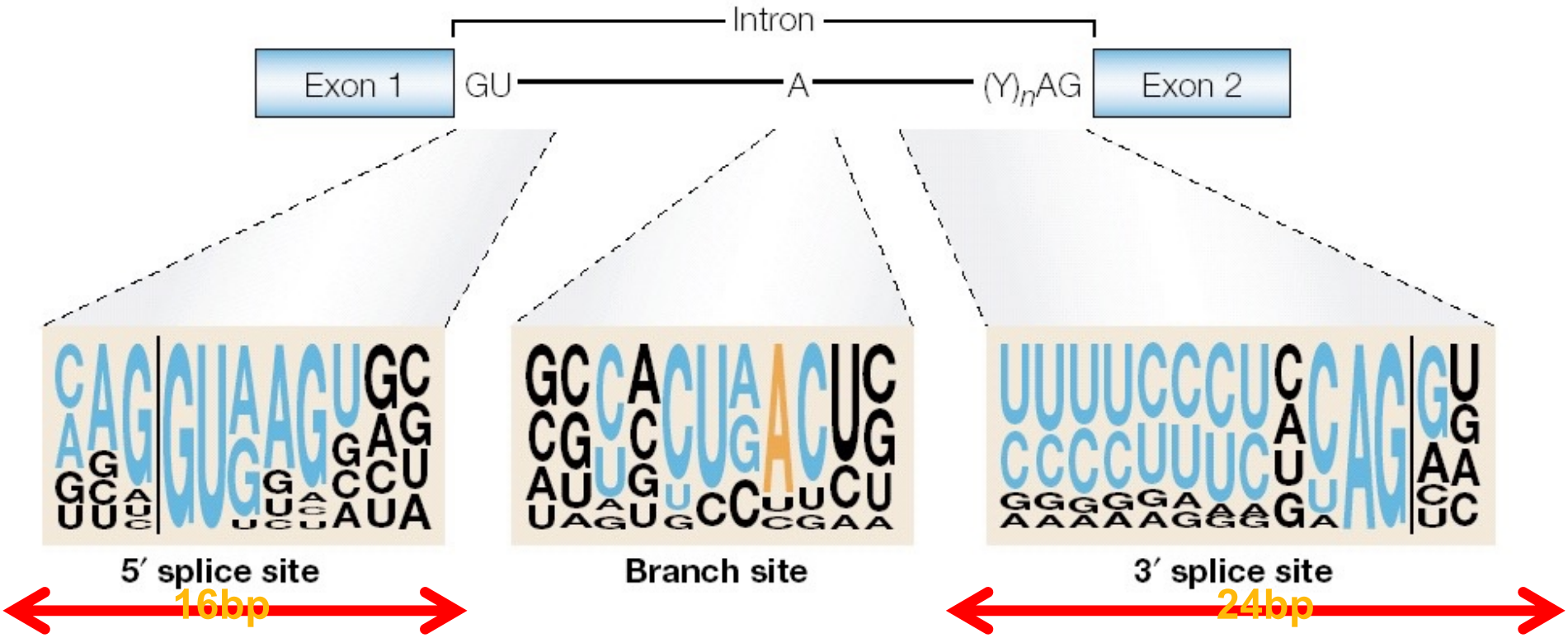



When the `score` is greater than some threshold (determined empirically to result in a desired sensitivity), we remember this position as being the potential site of a signal.

The most common signal sensor is the Position Weight Matrix:

| | | | | | | |
|---|---|---|---|---|---|---|
| A = 31%<br>T = 28%<br>C = 21%<br>G = 20% | A = 18%<br>T = 32%<br>C = 24%<br>G = 26% | **A**<br>100% | **T**<br>100% | **G**<br>100% | A = 19%<br>T = 20%<br>C = 29%<br>G = 32% | A = 24%<br>T = 18%<br>C = 26%<br>G = 32% |

# Splice site prediction

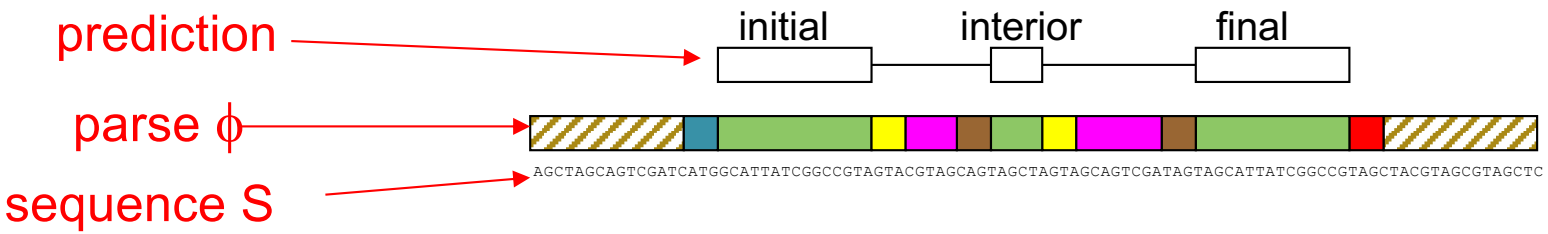



The splice site score is a combination of:

- first or second order inhomogeneous Markov models on windows around the acceptor and donor sites
- Maximal dependence decomposition (MDD) decision trees
- longer Markov models to capture difference between coding and non-coding on opposite sides of site (optional)
- maximal splice site score within 60 bp (optional)

# Gene Prediction with a GHMM

Given a sequence S, we would like to determine the parse $\phi$ of that sequence which segments the DNA into the most likely exon/intron structure:




The parse $\phi$ consists of the coordinates of the predicted exons, and corresponds to the precise sequence of states during the operation of the GHMM (and their duration, which equals the number of symbols each state emits).

This is the same as in an HMM except that in the HMM each state emits bases with fixed probability, whereas in the GHMM each state emits an entire feature such as an exon or intron.

*Bioinformatics*, 36(22-23), 2020, 5291–5298
doi: 10.1093/bioinformatics/btaa1044
Advance Access Publication Date: 16 December 2020
Original Paper

OXFORD

Genome analysis

# Helixer: cross-species gene annotation of large eukaryotic genomes using deep learning

**Felix Stiehler[1], Marvin Steinborn[1], Stephan Scholz, Daniela Dey[2], Andreas P. M. Weber[1] and Alisandra K. Denton [1,*]**

[1]Institue of Plant Biochemistry, Faculty of Mathematics and Natural Sciences, Heinrich-Heine-University, Dusseldorf 40225, Germany and [2]Institute of Human Genetics, Medical Faculty, RWTH Aachen University, Aachen 52062, Germany

*To whom correspondence should be addressed.

Associate Editor: Inanc Birol

Received on July 31, 2020; revised on November 6, 2020; editorial decision on December 4, 2020; accepted on December 7, 2020

## Abstract

**Motivation:** Current state-of-the-art tools for the *de novo* annotation of genes in eukaryotic genomes have to be specifically fitted for each species and still often produce annotations that can be improved much further. The fundamental algorithmic architecture for these tools has remained largely unchanged for about two decades, limiting learning capabilities. Here, we set out to improve the cross-species annotation of genes from DNA sequence alone with the help of deep learning. The goal is to eliminate the dependency on a closely related gene model while also improving the predictive quality in general with a fundamentally new architecture.

**Results:** We present Helixer, a framework for the development and usage of a cross-species deep learning model that improves significantly on performance and generalizability when compared to more traditional methods. We evaluate our approach by building a single vertebrate model for the base-wise annotation of 186 animal genomes and a separate land plant model for 51 plant genomes. Our predictions are shown to be much less sensitive to the length of the genome than those of a current state-of-the-art tool. We also present two novel post-processing techniques that each worked to further strengthen our annotations and show in-depth results of an RNA-Seq based comparison of our predictions. Our method does not yet produce comprehensive gene models but rather outputs base pair wise probabilities.

**Availability and implementation:** The source code of this work is available at https://github.com/weberlab-hhu/Helixer under the GNU General Public License v3.0. The trained models are available at https://doi.org/10.5281/zenodo.3974409

**Contact:** alisandra.denton@hhu.de

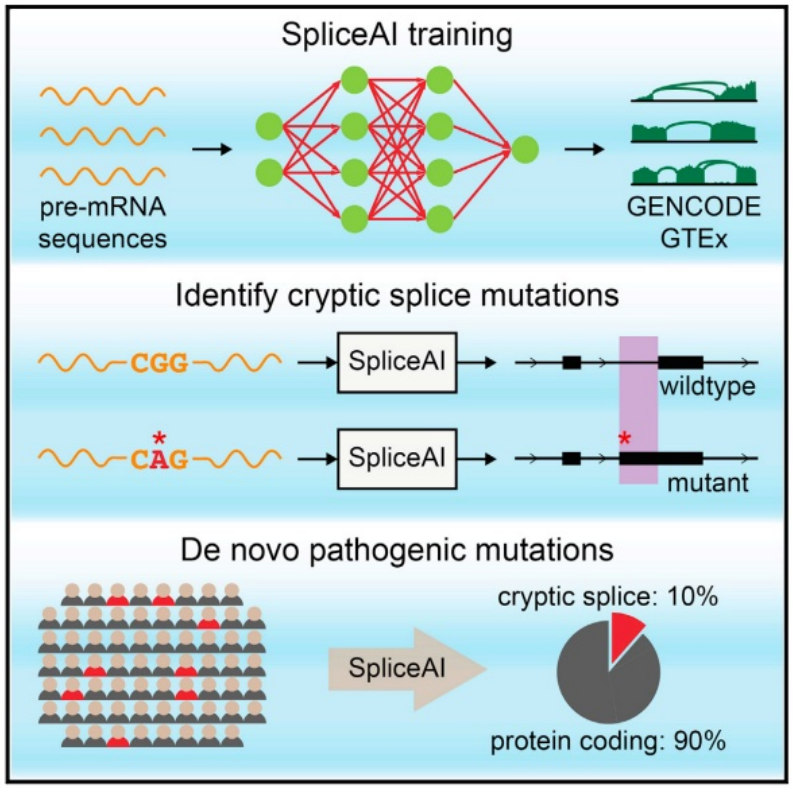
**Supplementary information:** Supplementary data are available at *Bioinformatics* online.

Cell



# Predicting Splicing from Primary Sequence with Deep Learning

## Graphical Abstract




## Highlights

- SpliceAI, a 32-layer deep neural network, predicts splicing from a pre-mRNA sequence
- 75% of predicted cryptic splice variants validate on RNA-seq
- Cryptic splicing may yield ~10% of pathogenic variants in neurodevelopmental disorders
- Cryptic splice variants frequently give rise to alternative splicing

## Authors

Kishore Jaganathan, Sofia Kyriazopoulou Panagiotopoulou, Jeremy F. McRae, ..., Serafim Batzoglou, Stephan J. Sanders, Kyle Kai-How Farh

## Correspondence

kfarh@illumina.com

## In Brief

A deep neural network precisely models mRNA splicing from a genomic sequence and accurately predicts noncoding cryptic splice mutations in patients with rare genetic diseases.

# Gene Finding Overview

- Prokaryotic gene finding distinguishes real genes and random ORFs
  - Prokaryotic genes have simple structure and are largely homogenous, making it relatively easy to recognize their sequence composition

- Eukaryotic gene finding identifies the genome-wide most probable gene models (set of exons)
  - "Probabilistic Graphical Model" to enforce overall gene structure, separate models to score splicing/transcription signals
  - Accuracy depends to a large extent on the quality of the training data



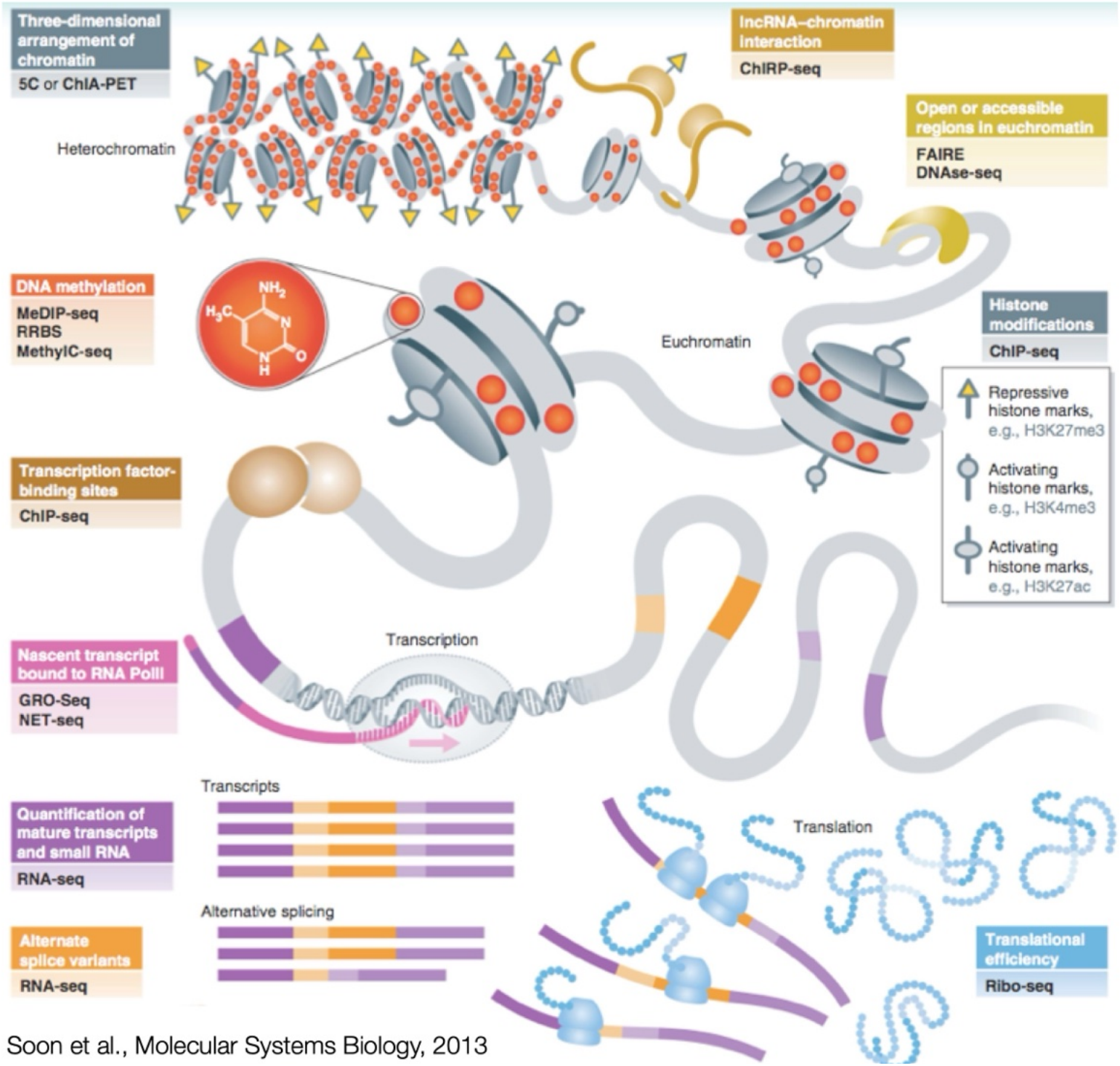
# Outline

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# Sequencing Assays

**The *Seq List (in chronological order)**

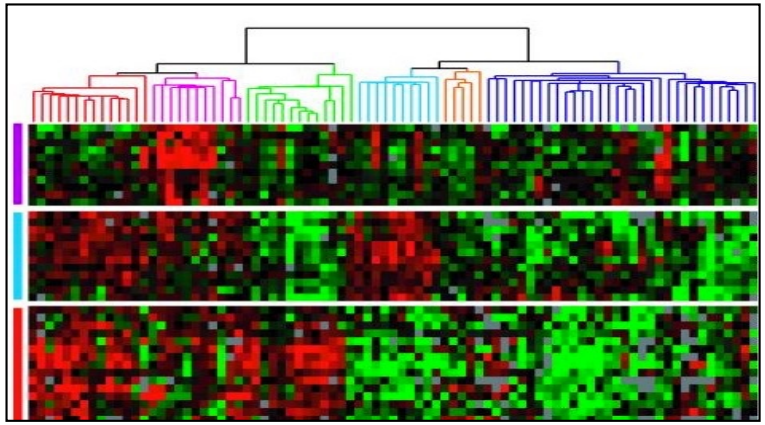

1. Gregory E. Crawford et al., "Genome-wide Mapping of DNase Hypersensitive Sites Using Massively Parallel Signature Sequencing (MPSS)," Genome Research 16, no. 1 (January 1, 2006): 123–131, doi:10.1101/gr.4074106.
2. David S. Johnson et al., "Genome-Wide Mapping of in Vivo Protein-DNA Interactions," Science 316, no. 5830 (June 8, 2007): 1497–1502, doi:10.1126/science.1141319.
3. Tarjei S. Mikkelsen et al., "Genome-wide Maps of Chromatin State in Pluripotent and Lineage-committed Cells," Nature 448, no. 7153 (August 2, 2007): 553–560, doi:10.1038/nature06008.
4. Thomas A. Down et al., "A Bayesian Deconvolution Strategy for Immunoprecipitation-based DNA Methylome Analysis," Nature Biotechnology 26, no. 7 (July 2008): 779–785, doi:10.1038/nbt1414.
5. Ali Mortazavi et al., "Mapping and Quantifying Mammalian Transcriptomes by RNA-Seq," Nature Methods 5, no. 7 (July 2008): 621–628, doi:10.1038/nmeth.1226.
6. Nathan A. Baird et al., "Rapid SNP Discovery and Genetic Mapping Using Sequenced RAD Markers," PLoS ONE 3, no. 10 (October 13, 2008): e3376, doi:10.1371/journal.pone.0003376.
7. Leighton J. Core, Joshua J. Waterfall, and John T. Lis, "Nascent RNA Sequencing Reveals Widespread Pausing and Divergent Initiation at Human Promoters," Science 322, no. 5909 (December 19, 2008): 1845–1848, doi:10.1126/science.1162228.
8. Chao Xie and Martti T. Tammi, "CNV-seq, a New Method to Detect Copy Number Variation Using High-throughput Sequencing," BMC Bioinformatics 10, no. 1 (March 6, 2009): 80, doi:10.1186/1471-2105-10-80.
9. Jay R. Hesselberth et al., "Global Mapping of protein-DNA Interactions in Vivo by Digital Genomic Footprinting," Nature Methods 6, no. 4 (April 2009): 283–289, doi:10.1038/nmeth.1313.
10. Nicholas T. Ingolia et al., "Genome-Wide Analysis in Vivo of Translation with Nucleotide Resolution Using Ribosome Profiling," Science 324, no. 5924 (April 10, 2009): 218–223, doi:10.1126/science.1168978.
11. Alayne L. Brunner et al., "Distinct DNA Methylation Patterns Characterize Differentiated Human Embryonic Stem Cells and Developing Human Fetal Liver," Genome Research 19, no. 6 (June 1, 2009): 1044–1056, doi:10.1101/gr.088773.108.
12. Mayumi Oda et al., "High-resolution Genome-wide Cytosine Methylation Profiling with Simultaneous Copy Number Analysis and Optimization for Limited Cell Numbers," Nucleic Acids Research 37, no. 12 (July 1, 2009): 3829–3839, doi:10.1093/nar/gkp260.
13. Zachary D. Smith et al., "High-throughput Bisulfite Sequencing in Mammalian Genomes," Methods 48, no. 3 (July 2009): 226–232, doi:10.1016/j.ymeth.2009.05.003.
14. Andrew M. Smith et al., "Quantitative Phenotyping via Deep Barcode Sequencing," Genome Research (July 21, 2009).



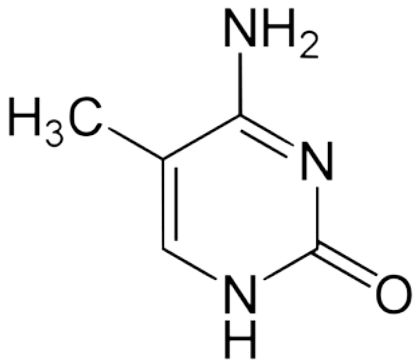

Soon et al., Molecular Systems Biology, 2013

# *-seq in 4 short vignettes

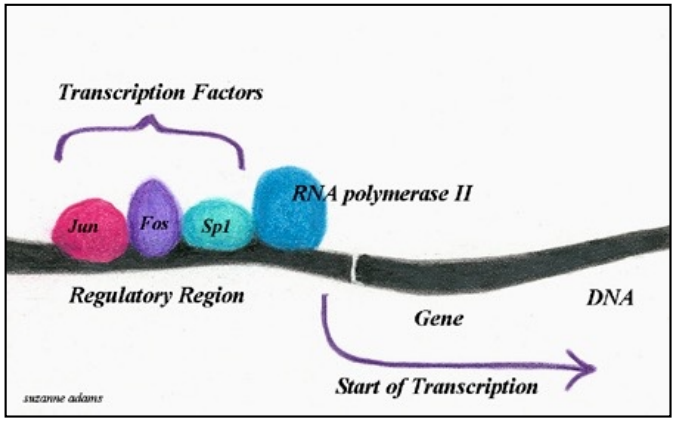
## RNA-seq



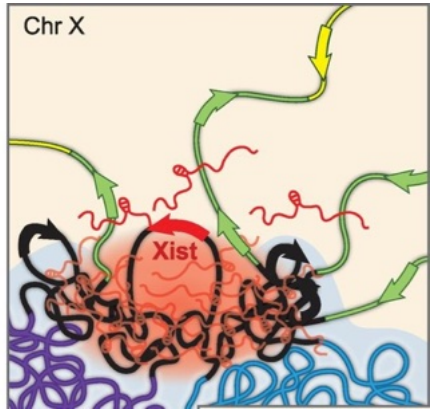
## Methyl-seq



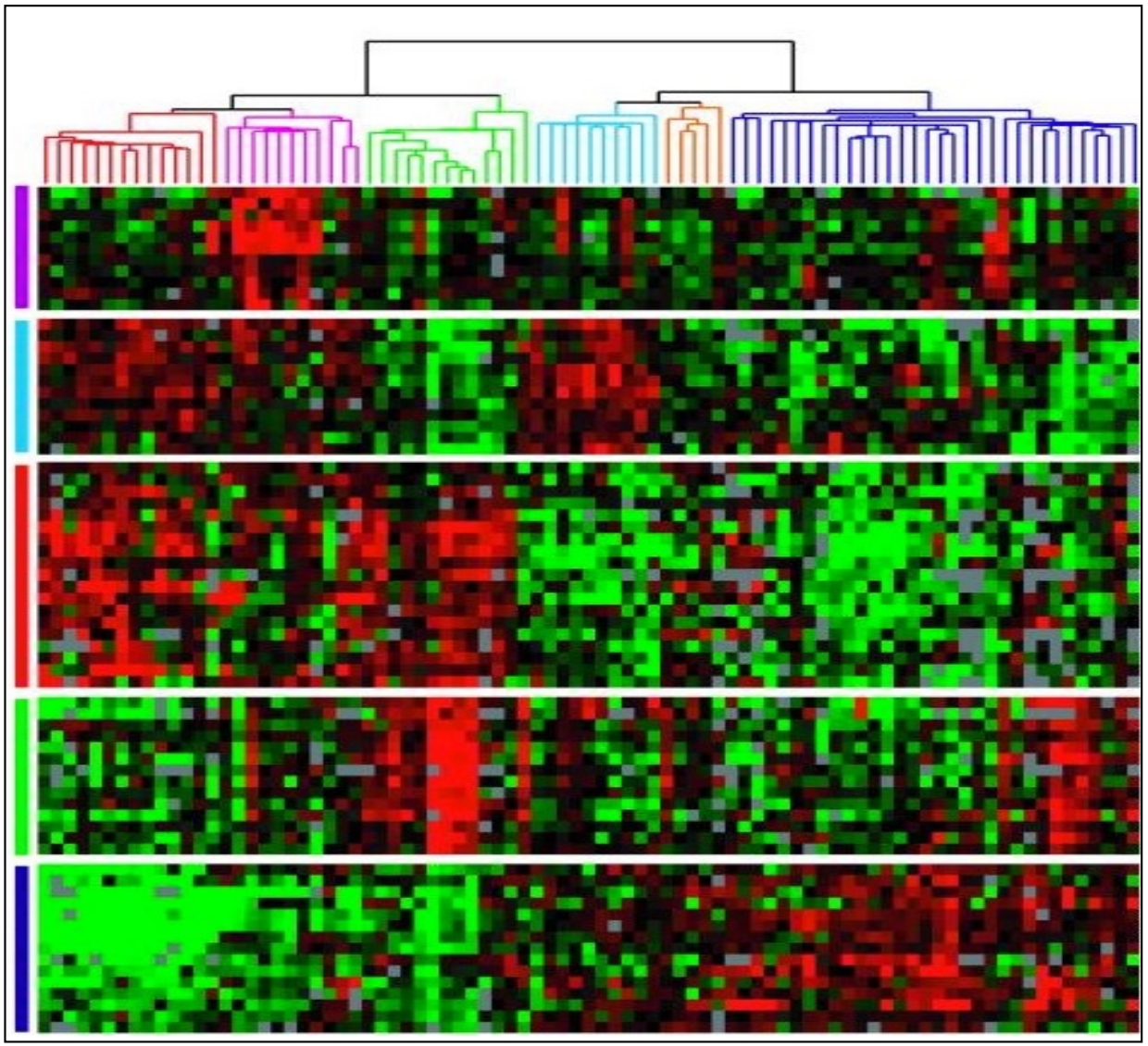

## ChIP-seq




## Hi-C

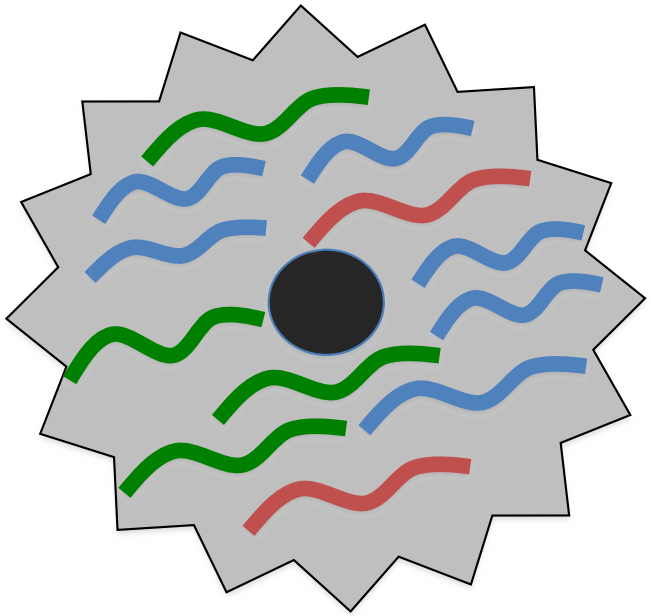


# RNA-seq



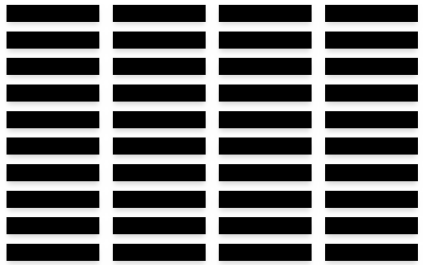
**Gene expression patterns of breast carcinomas distinguish tumor subclasses with clinical implications.**
Sørlie et al (2001) *PNAS*. 98(19):10869-74.

# RNA-seq Overview



Sequencing

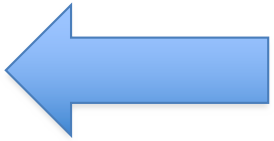




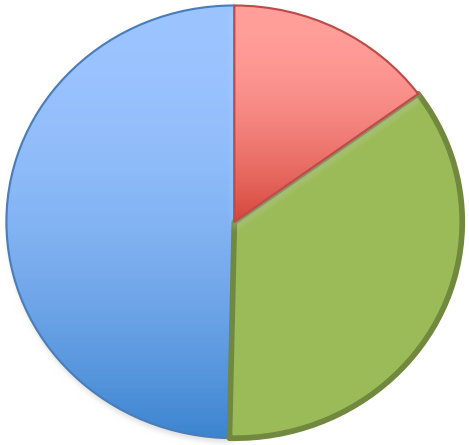


Mapping
& Assembly

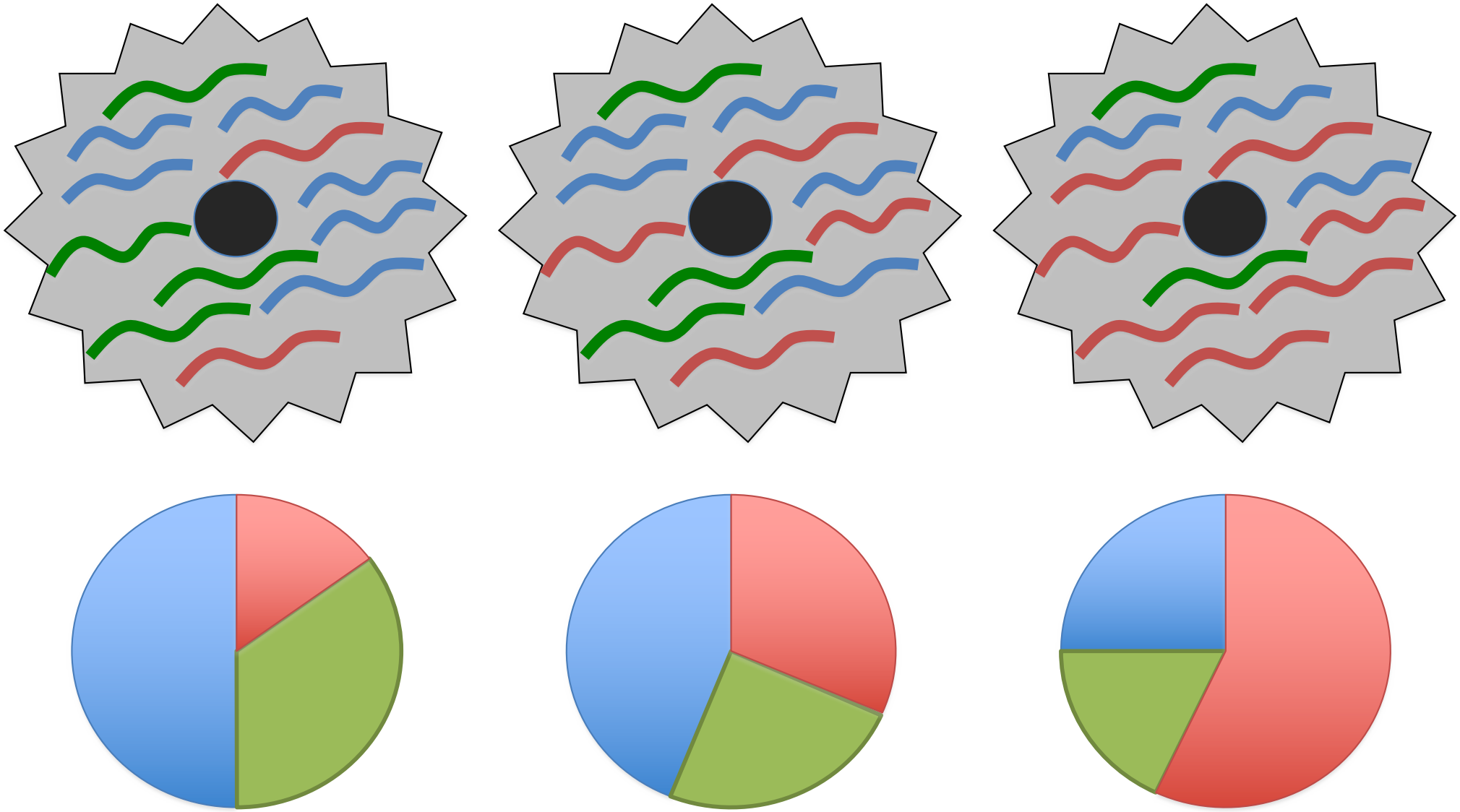




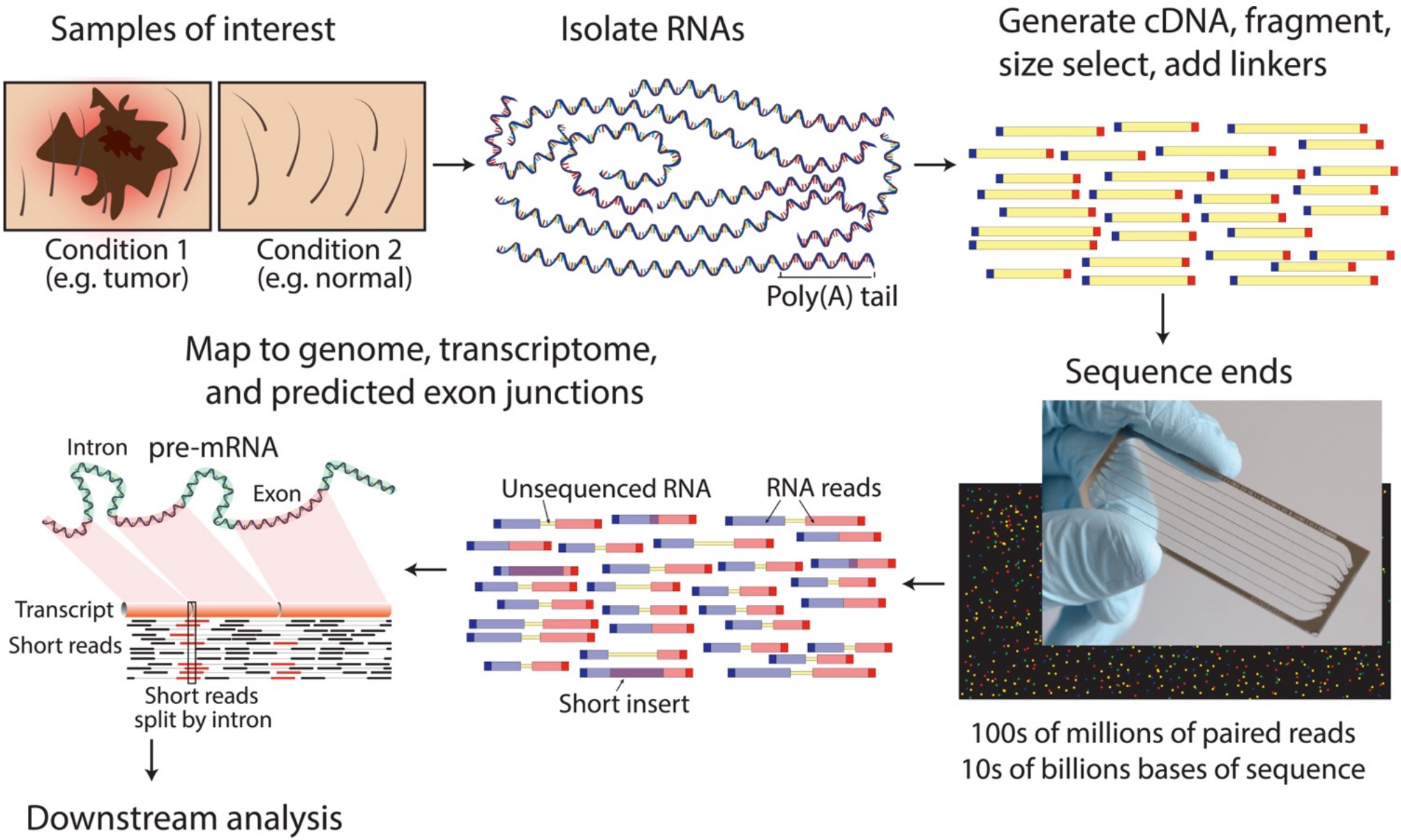
Quantification



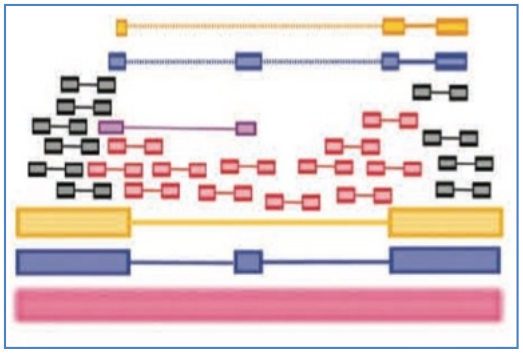
# RNA-seq Overview



# RNA-seq Overview



# RNA-seq Challenges



**Challenge 1: Eukaryotic genes are spliced**

# RNA-Seq Approaches

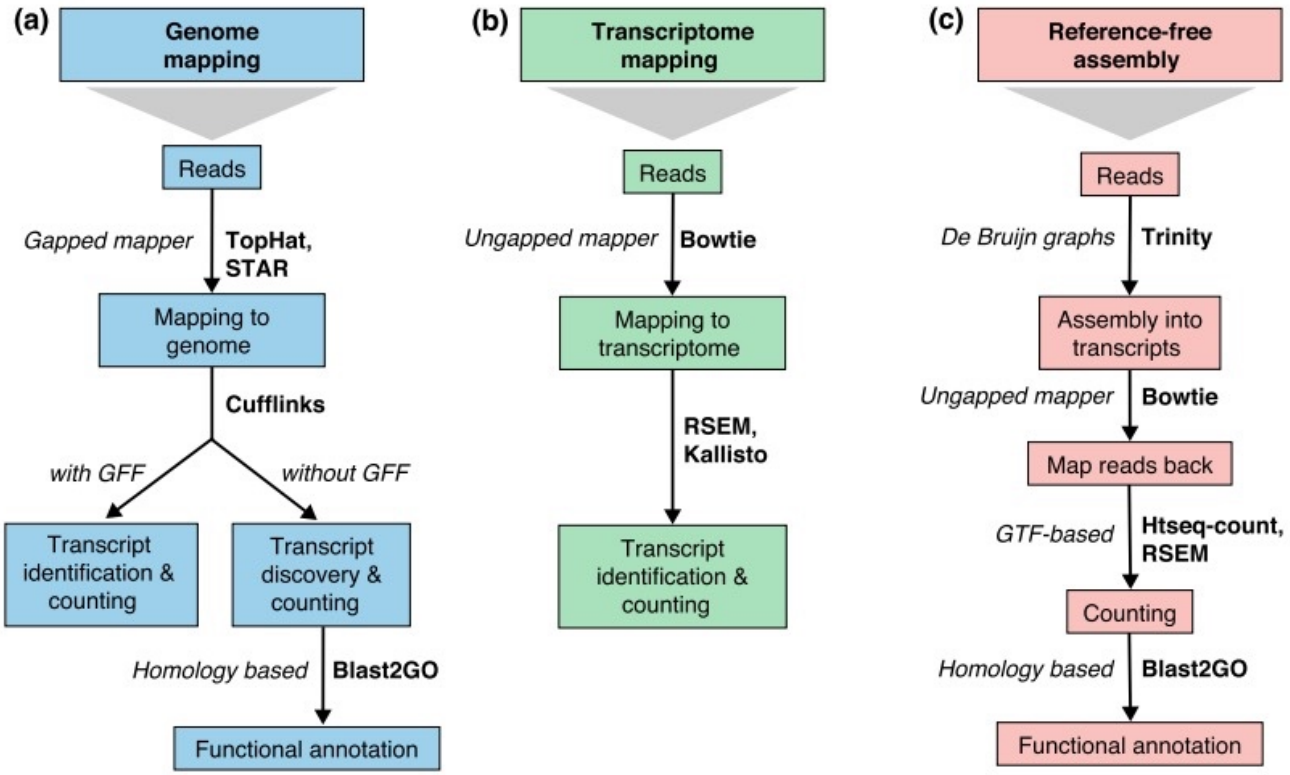



**Fig. 2** Read mapping and transcript identification strategies. Three basic strategies for regular RNA-seq analysis. **a** An annotated genome is available and reads are mapped to the genome with a gapped mapper. Next (novel) transcript discovery and quantification can proceed with or without an annotation file. Novel transcripts are then functionally annotated. **b** If no novel transcript discovery is needed, reads can be mapped to the reference transcriptome using an ungapped aligner. Transcript identification and quantification can occur simultaneously. **c** When no genome is available, reads need to be assembled first into contigs or transcripts. For quantification, reads are mapped back to the novel reference transcriptome and further analysis proceeds as in (**b**) followed by the functional annotation of the novel transcripts as in (**a**). Representative software that can be used at each analysis step are indicated in *bold text*. Abbreviations: *GFF* General Feature Format, *GTF* gene transfer format, *RSEM* RNA-Seq by Expectation Maximization

***A survey of best practices for RNA-seq data analysis***
Conesa et al (2016) Genome Biology. doi 10.1186/s13059-016-0881-8

# RNA-Seq Approaches

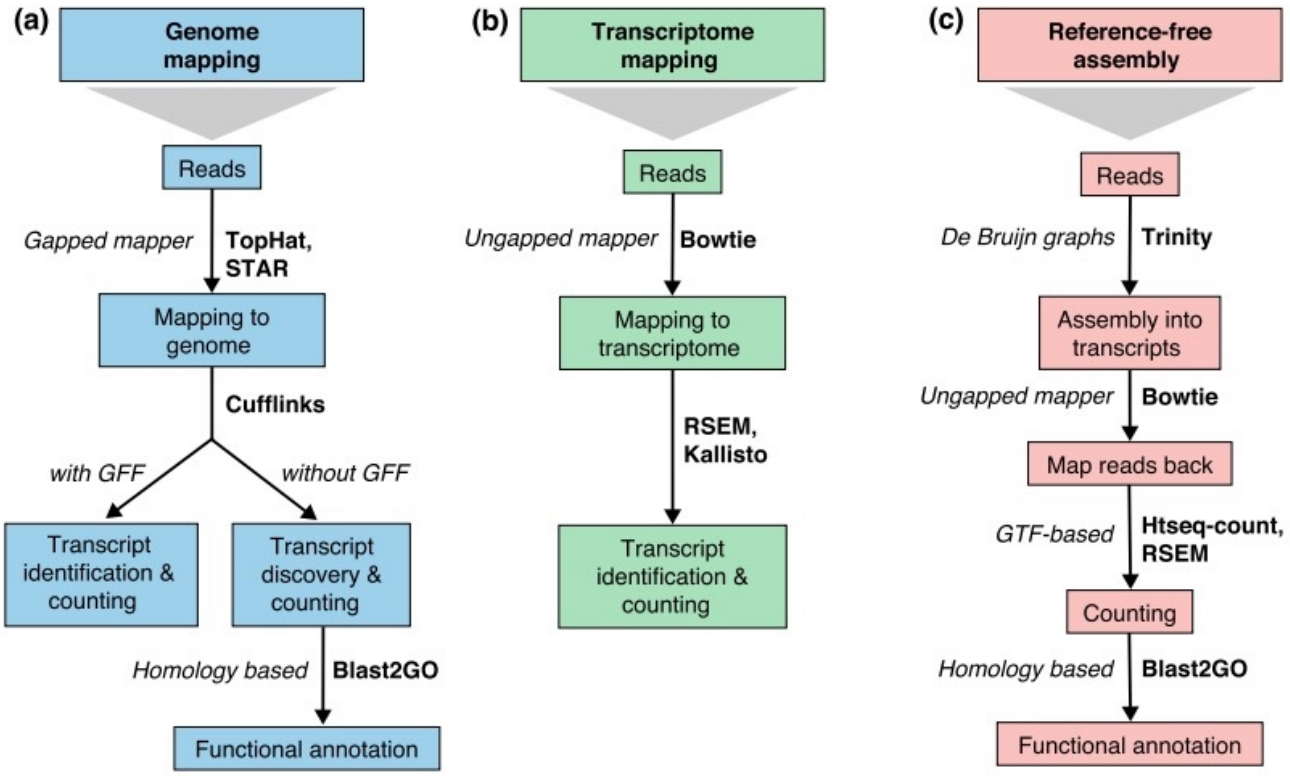



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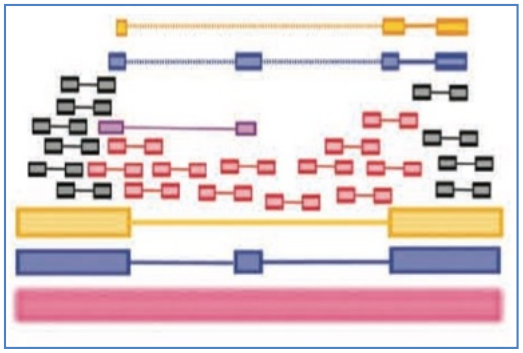
Which approach should we use?

It depends….

***A survey of best practices for RNA-seq data analysis***

Conesa et al (2016) Genome Biology. doi 10.1186/s13059-016-0881-8

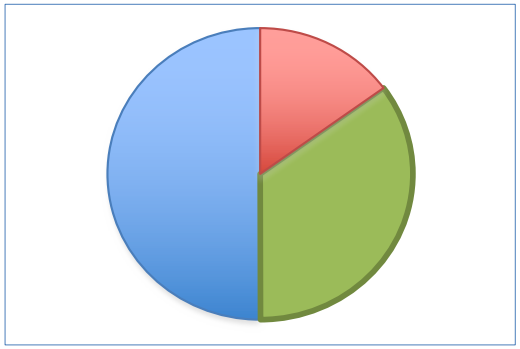
# RNA-seq Challenges



**Challenge 1: Eukaryotic genes are spliced**

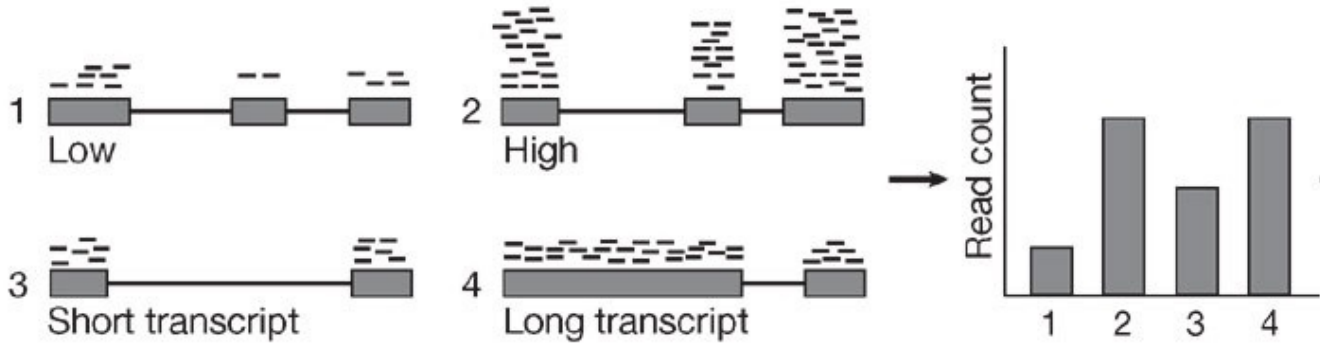
Solution: Use a spliced aligner, and assemble isoforms

**TopHat: discovering spliced junctions with RNA-Seq.**
Trapnell et al (2009) *Bioinformatics*. 25:0 1105-1111



**Challenge 2: Read Count != Transcript abundance**

# RPKM, FPKM, TPM

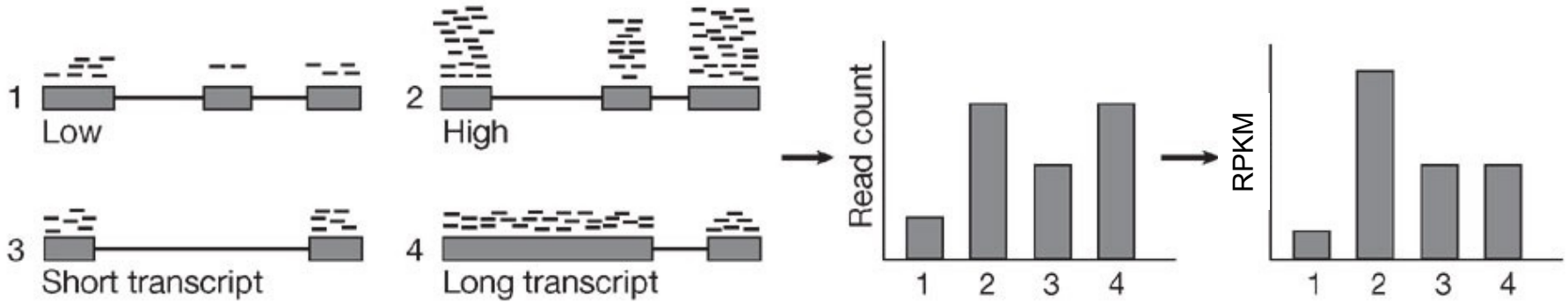



***Counting Reads that align to a gene DOESN'T work!***

- Overall Coverage: 1M reads in experiment 1 vs 10M reads in experiment 2
- Gene Length: gene 3 is 10kbp, gene 4 is 100kbp

***1. RPKM: Reads Per Kilobase of Exon Per Million Reads Mapped (Mortazavi et al, 2008)***

# RPKM, FPKM, TPM




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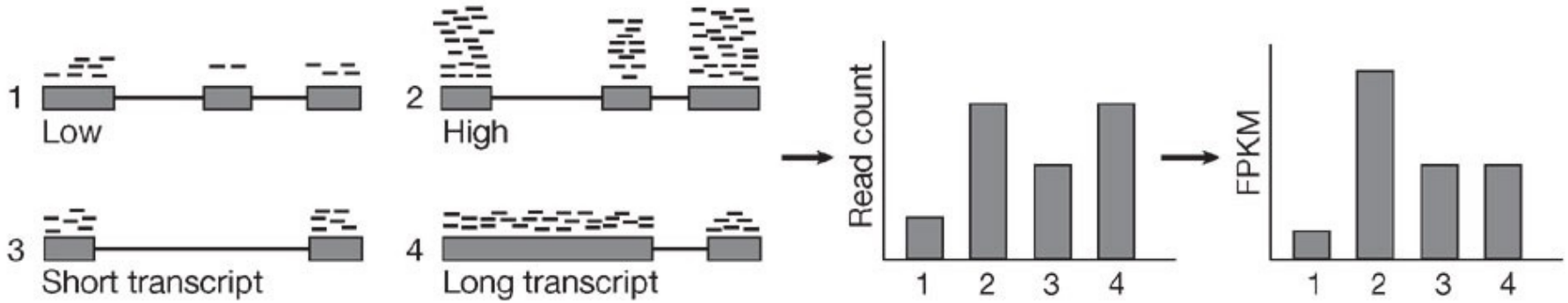
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***1. RPKM: Reads Per Kilobase of Exon Per Million Reads Mapped (Mortazavi et al, 2008)***

(Count reads aligned to gene) / (length of gene in kilobases) / (# millions of read mapped)

=> Wait a second, reads in a pair arent independent!

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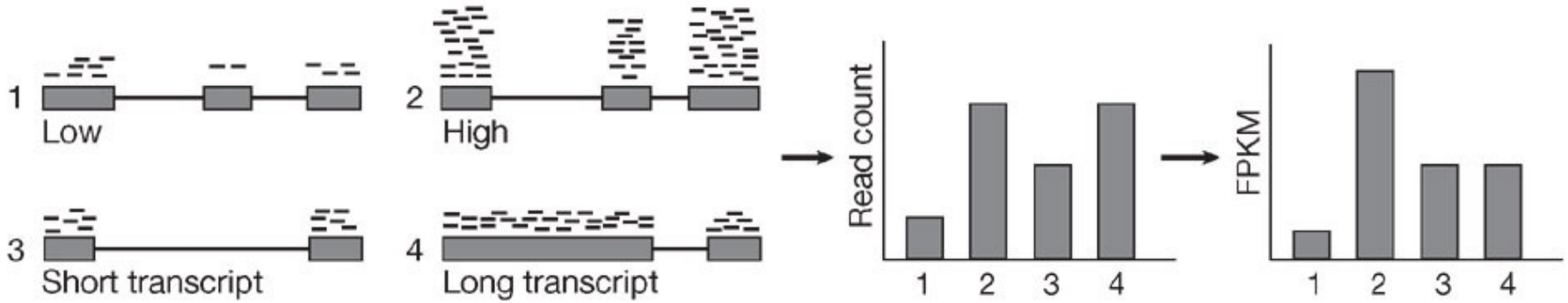
=> Wait a second, reads in a pair arent independent!

***2. FPKM: Fragments Per Kilobase of Exon Per Million Reads Mapped (Trapnell et al, 2010)***

$\Rightarrow$ Does a much better job with short exons & short genes by boosting coverage

$\Rightarrow$ Wait a second, FPKM depends on the average transcript length!

# RPKM, FPKM, TPM




***Counting Reads that align to a gene DOESN'T work!***

- Overall Coverage: 1M reads in experiment 1 vs 10M reads in experiment 2
- Gene Length: gene 3 is 10kbp, gene 4 is 100kbp

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=> Wait a second, reads in a pair arent independent!

***2. FPKM: Fragments Per Kilobase of Exon Per Million Reads Mapped (Trapnell et al, 2010)***

=> Wait a second, FPKM depends on the average transcript length!

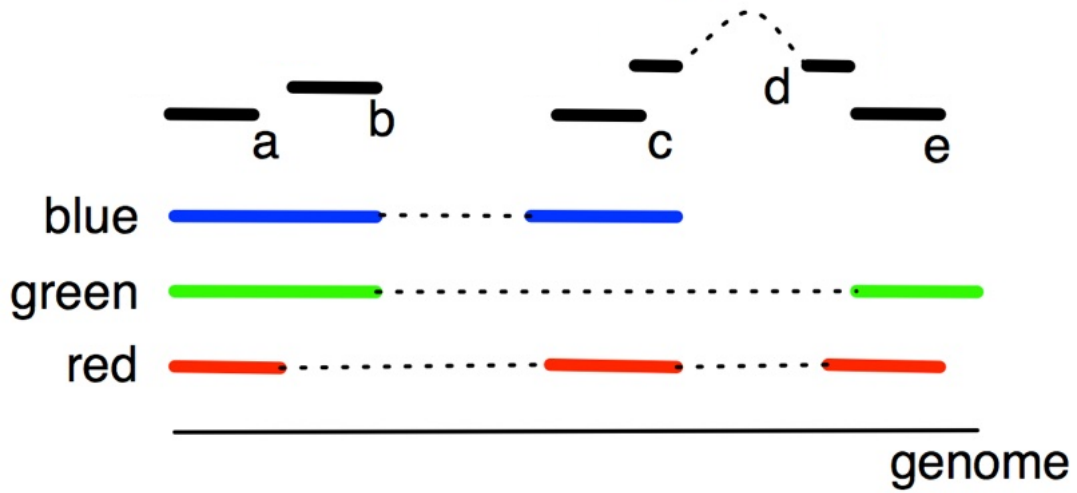
***3. TPM: Transcripts Per Million (Li et al, 2011)***

⇒ If you were to sequence one million full length transcripts, TPM is the number of transcripts you would have seen of type i, given the abundances of the other transcripts in your sample

=> Recommend you use TPM for all analysis, easy to compute given FPKM

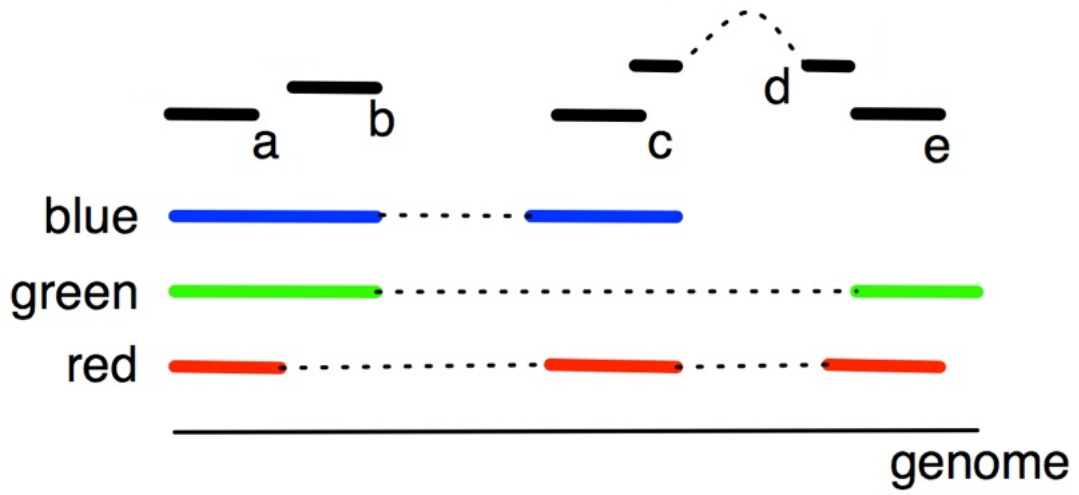
$$\mathrm{TPM}_i = \left( \frac{\mathrm{FPKM}_i}{\sum_j \mathrm{FPKM}_j} \right) \cdot 10^6$$

# Multi-mapping? Isoform ambiguity? Expectation Maximization to the Rescue




***Models for transcript quantification from RNA-seq***
Pachter, L (2011) arXiv. 1104.3889 [q-bio.GN]

# Multi-mapping? Isoform ambiguity? Expectation Maximization to the Rescue




The gene has three isoforms (red, green, blue) of the same length.
Our initial expectation is all 3 isoforms are equally expressed

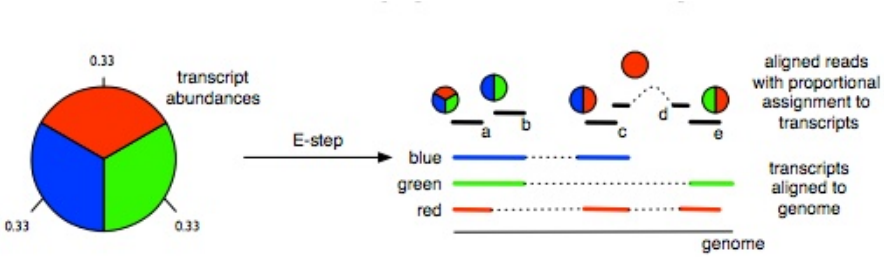
There are five reads (a,b,c,d,e) mapping to the gene.

- Read a maps to all three isoforms
- Read d only to red
- Reads b,c,e map to each of the three pairs of isoforms.

What is the most likely expression level of each isoform?

***Models for transcript quantification from RNA-seq***
Pachter, L (2011) arXiv. 1104.3889 [q-bio.GN]

# Multi-mapping? Isoform ambiguity? Expectation Maximization to the Rescue



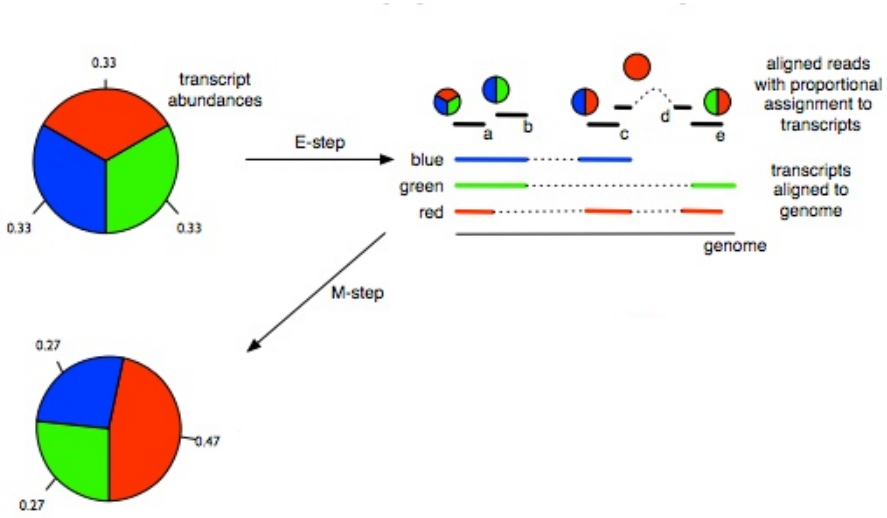

The gene has three isoforms (red, green, blue) of the same length. Initially every isoform is assigned the same abundance (red=1/3, green=1/3, blue=1/3)

There are five reads (a,b,c,d,e) mapping to the gene. Read a maps to all three isoforms, read d only to red, and the other three (reads b,c,e) to each of the three pairs of isoforms.

During the expectation (E) step reads are proportionately assigned to transcripts according to the (current) isoform abundances (RGB): a=(.33,.33,.33), b=(0,.5,.5), c=(.5,.5), d=(1,0,0), e=(.5,.5,0)

***Models for transcript quantification from RNA-seq***
Pachter, L (2011) arXiv. 1104.3889 [q-bio.GN]

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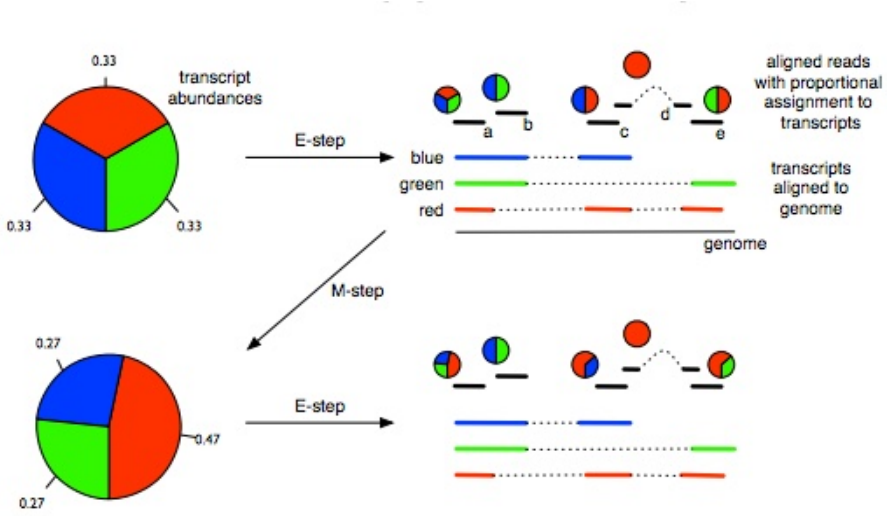
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Next, during the maximization (M) step isoform abundances are recalculated from the proportionately assigned read counts:
red: 0.47 = (0.33 + 0.5 + 1 + 0.5)/(2.33 + 1.33 + 1.33)
blue: 0.27 = (0.33 + 0.5 + 0.5)/(2.33 + 1.33 + 1.33)
green: 0.27 = (0.33 + 0.5 + 0.5)/(2.33 + 1.33 + 1.33)

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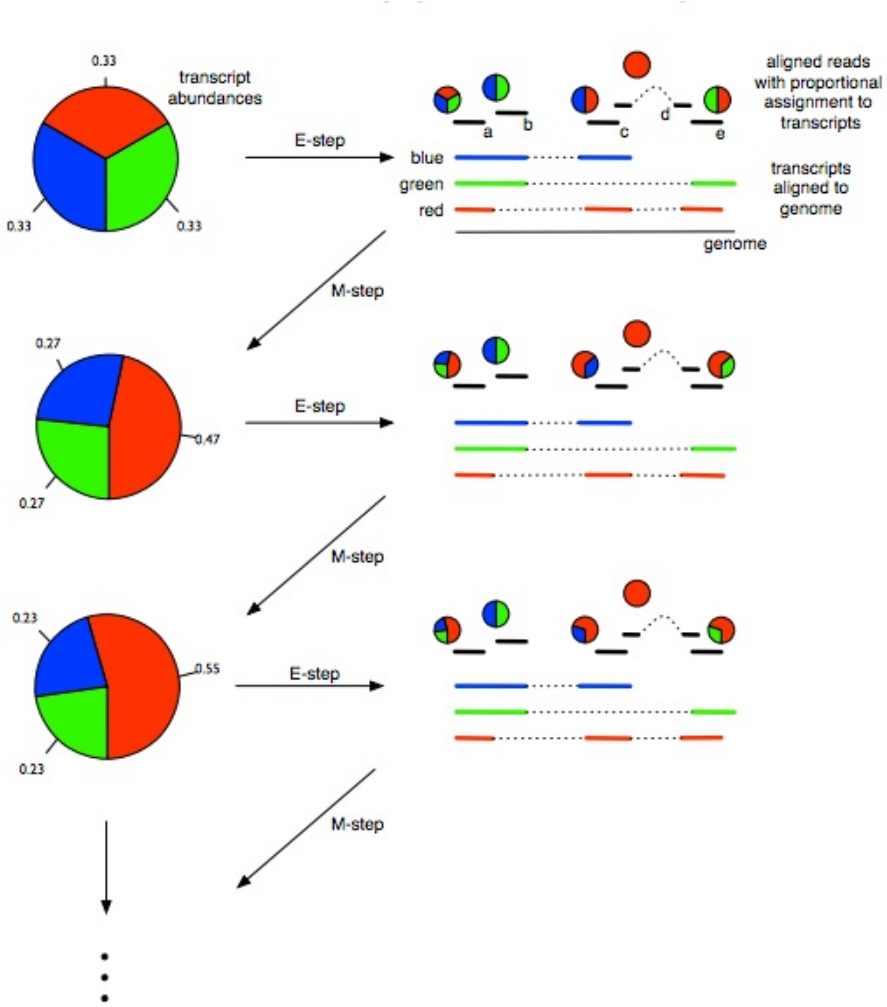
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Repeat until convergence!

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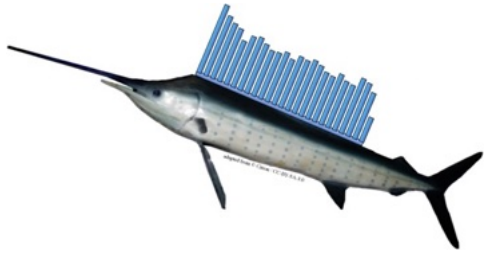
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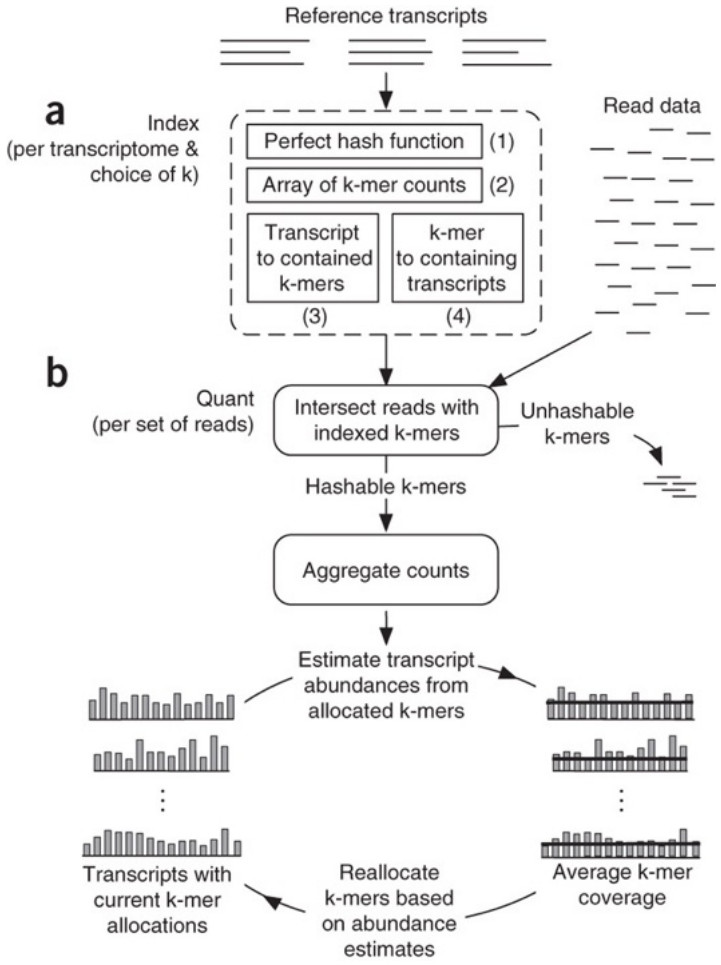
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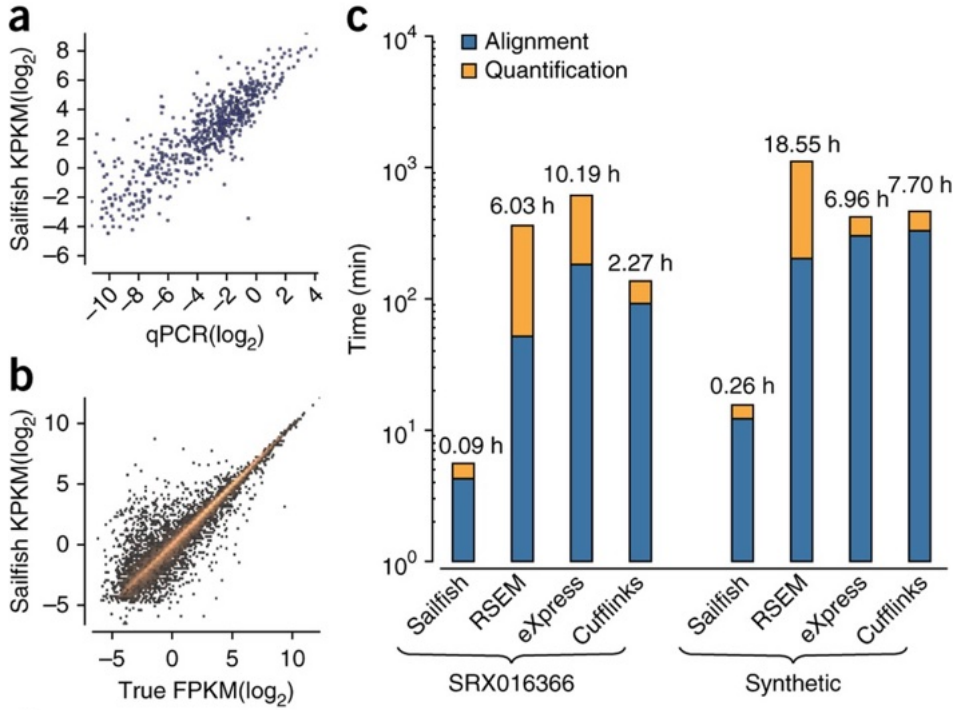
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***Models for transcript quantification from RNA-seq***
Pachter, L (2011) arXiv. 1104.3889 [q-bio.GN]

# Sailfish: Fast & Accurate RNA-seq Quantification





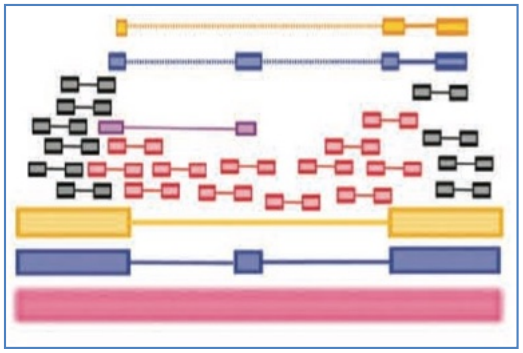




d

| | Human brain tissue | | | | Synthetic | | | |
|---|---|---|---|---|---|---|---|---|
| | Sailfish | RSEM | eXpress | Cufflinks | Sailfish | RSEM | eXpress | Cufflinks |
| Pearson | 0.85 | 0.82 | 0.85 | 0.85 | 0.96 | 0.96 | 0.95 | 0.94 |
| Spearman | 0.84 | 0.80 | 0.85 | 0.85 | 0.76 | 0.77 | 0.77 | 0.75 |
| RMSE | – | – | – | – | 6.06 | 8.90 | 8.83 | 10.05 |
| medPE | – | – | – | – | 6.50 | 12.48 | 14.06 | 12.42 |

***Sailfish enables alignment-free isoform quantification from RNA-seq reads using lightweight algorithms***
Patro et al (2014) Nature Biotechnology 32, 462–464 doi:10.1038/nbt.2862

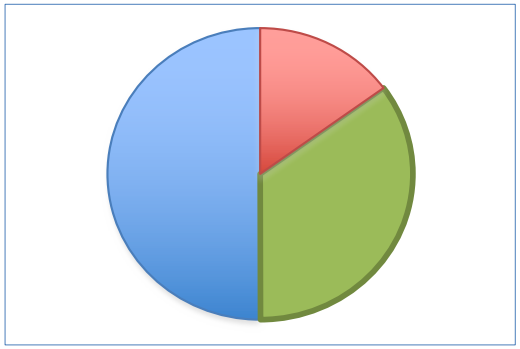
# RNA-seq Challenges



## Challenge 1: Eukaryotic genes are spliced

Solution: Use a spliced aligner, and assemble isoforms

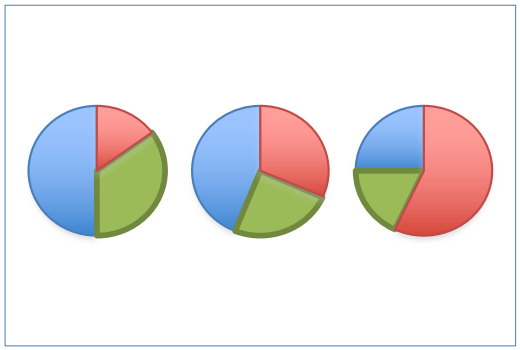
**TopHat: discovering spliced junctions with RNA-Seq.**
Trapnell et al (2009) *Bioinformatics*. 25:0 1105-1111



## Challenge 2: Read Count != Transcript abundance

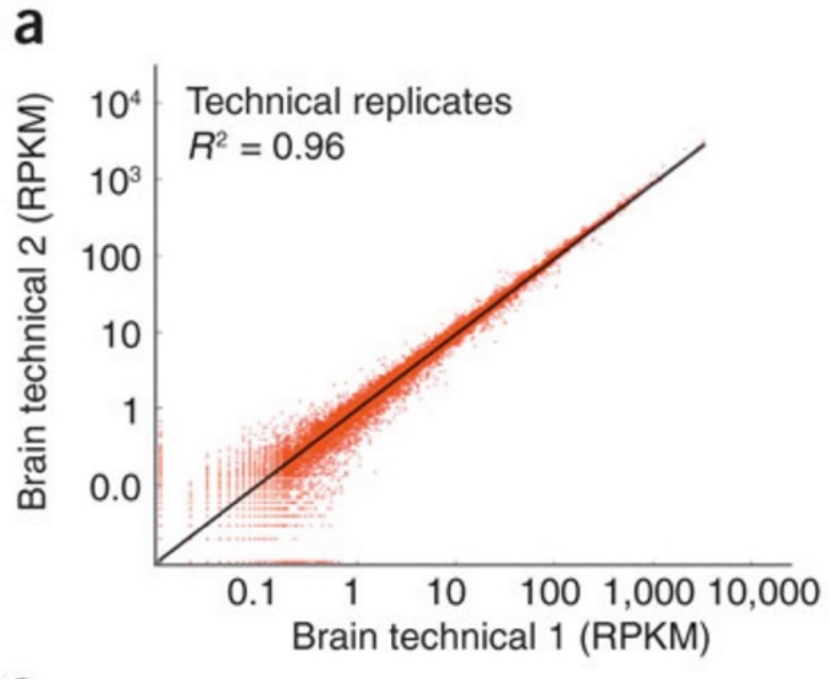
Solution: Infer underlying abundances (e.g. TPM)

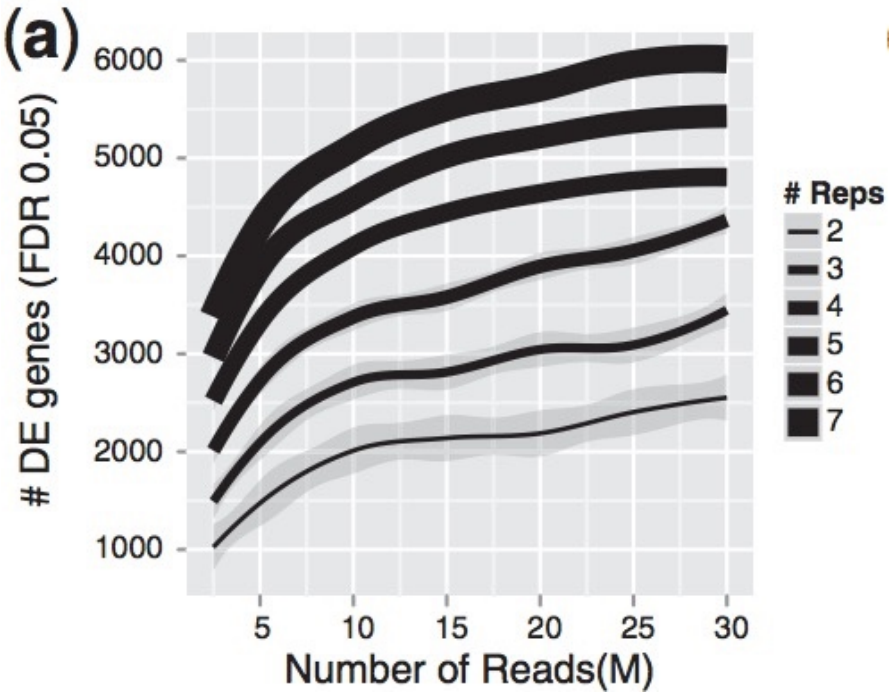
**Transcript assembly and quantification by RNA-seq**
Trapnell et al (2010) *Nat. Biotech.* 25(5): 511-515



## Challenge 3: Transcript abundances are stochastic

# How Many Replicates?



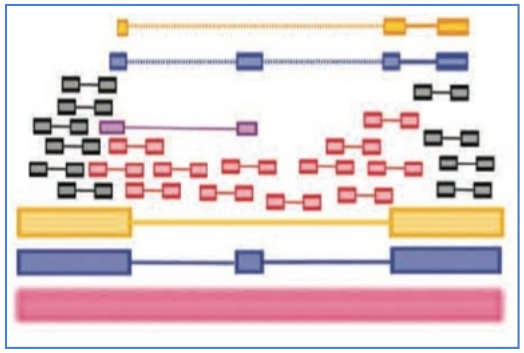




Why don't we have perfect replicates?

***Mapping and quantifying mammalian transcriptomes by RNA-Seq***
Mortazavi et al (2008) Nature Methods. 5, 62-628

***RNA-seq differential expression studies: more sequence or more replication?***
Liu et al (2013) Bioinformatics. doi:10.1093/bioinformatics/btt688

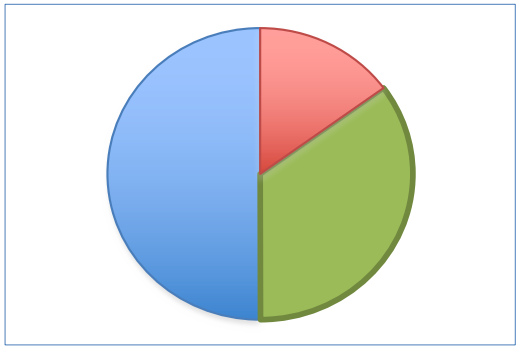
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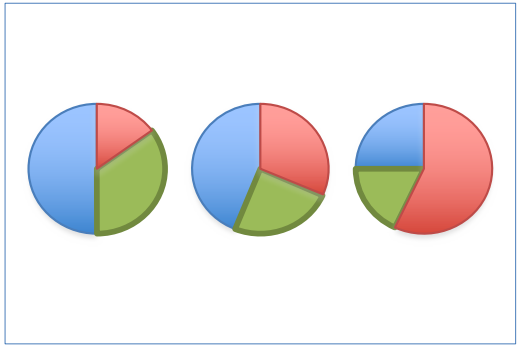
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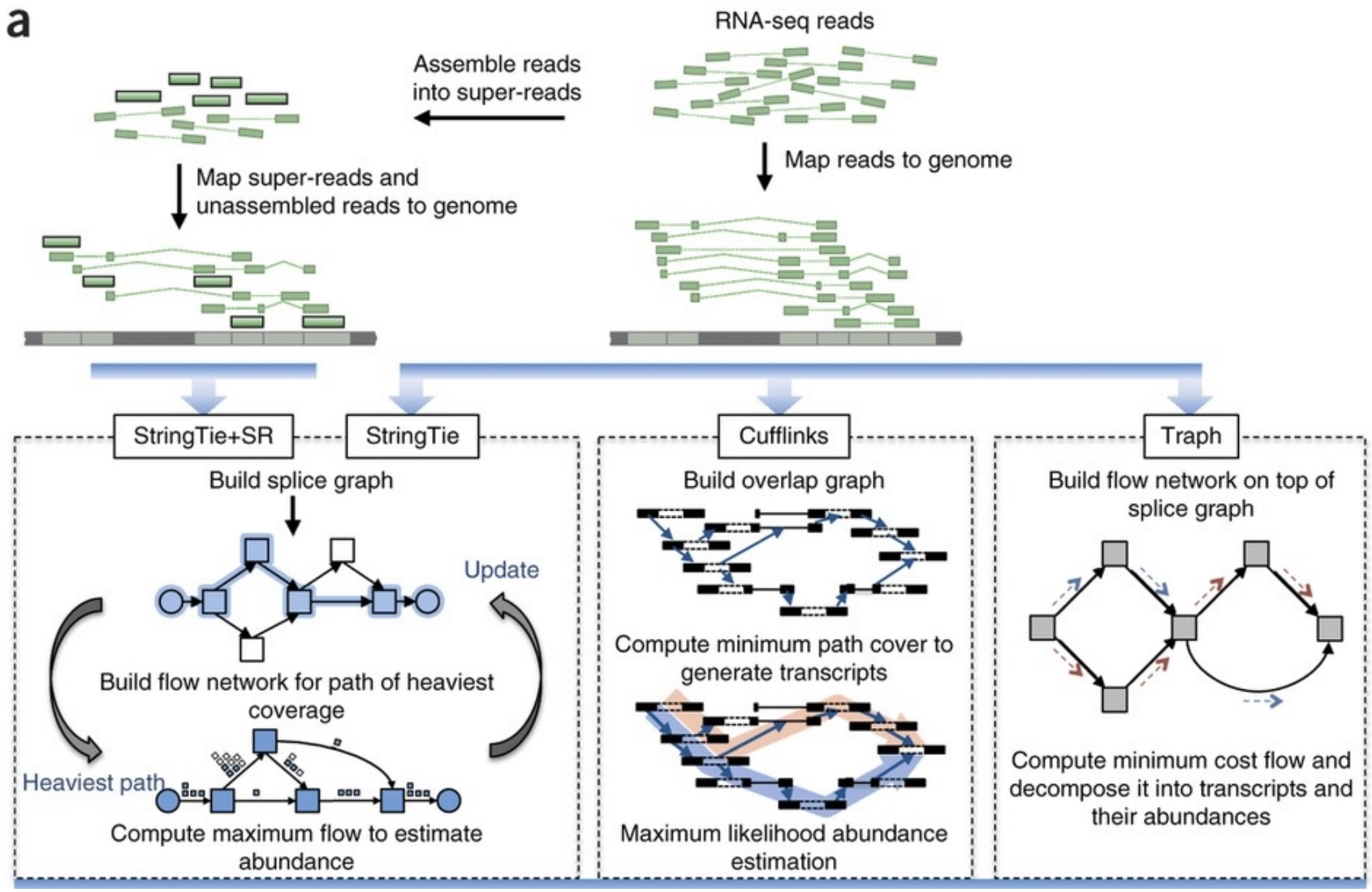


## Challenge 3: Transcript abundances are stochastic

Solution: Replicates, replicates, and more replicates

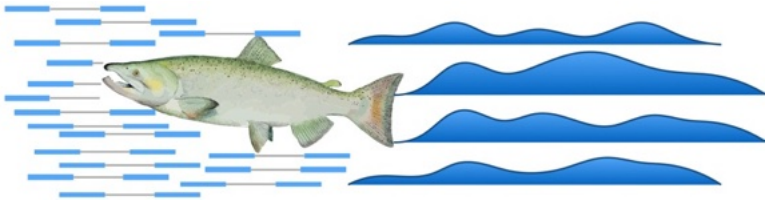
**RNA-seq differential expression studies: more sequence or more replication?**
Liu et al (2013) *Bioinformatics*. doi:10.1093/bioinformatics/btt688

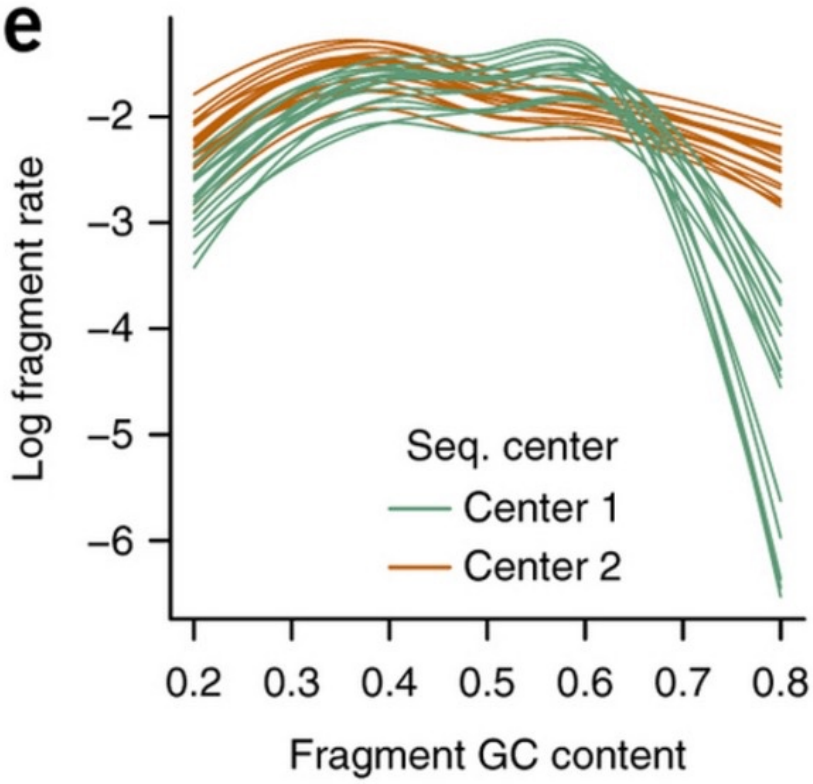
# Isoform Quantification Approaches

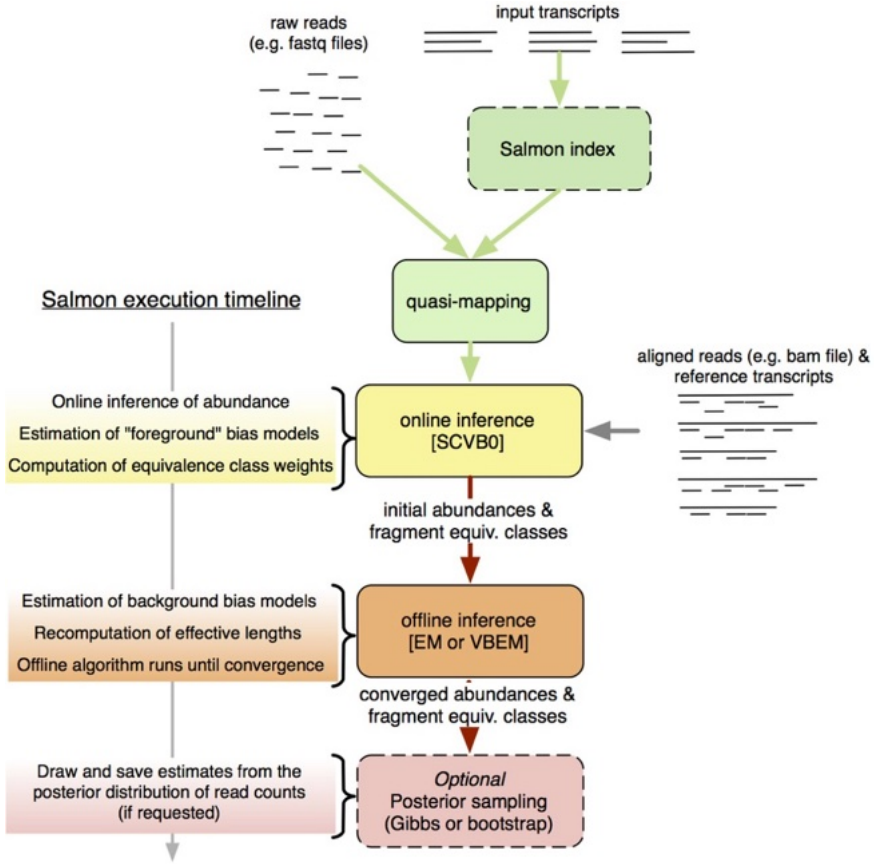



***StringTie enables improved reconstruction of a transcriptome from RNA-seq reads.***
Pertea M, et al. (2015) Nature Biotechnology. doi: 10.1038/nbt.3122.

# Salmon: The ultimate RNA-seq Pipeline?



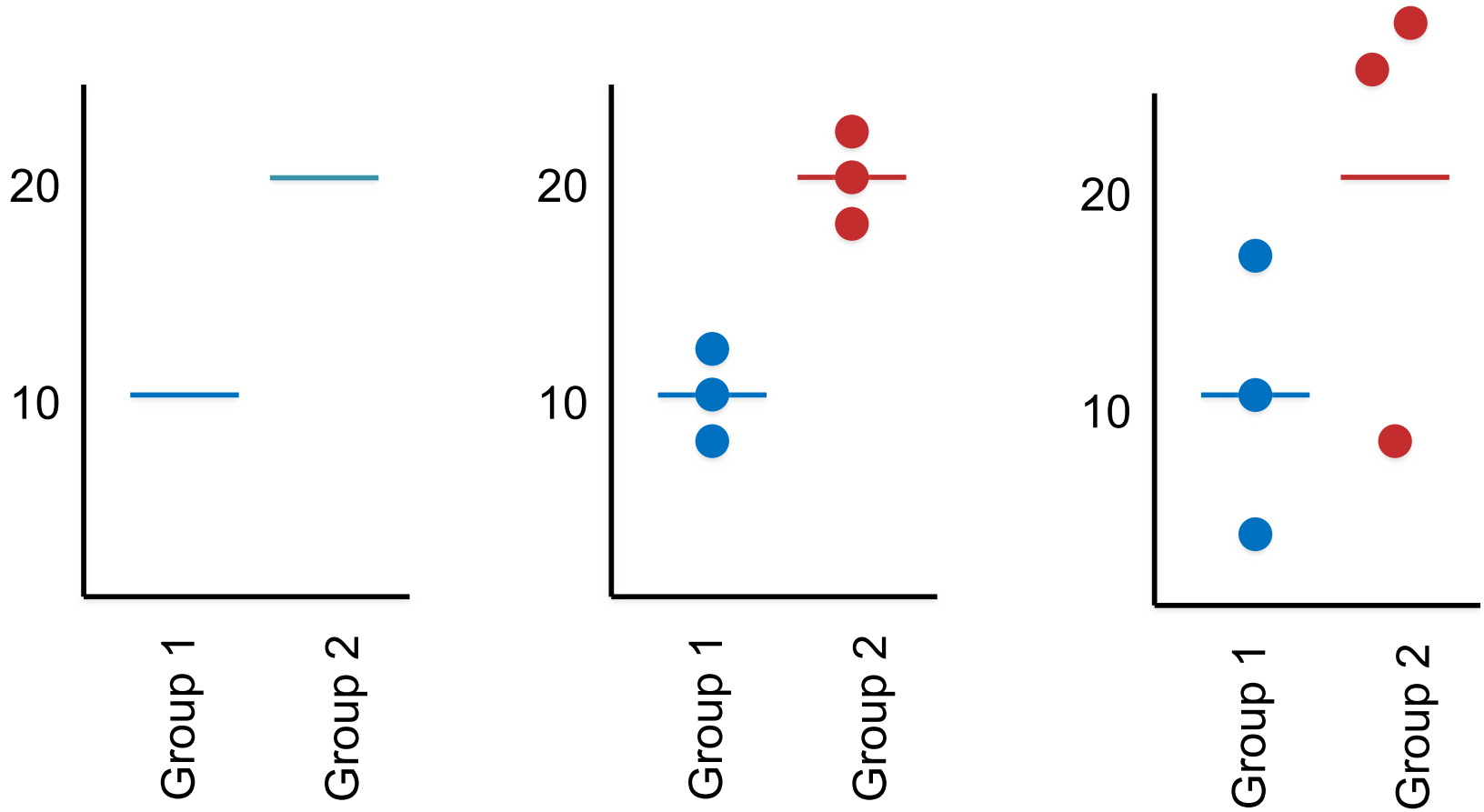



***Modeling of RNA-seq fragment sequence bias reduces systematic errors in transcript abundance estimation***
Love et al (2016) Nature Biotechnology 34, 1287–1291 (2016) doi:10.1038/nbt.3682

***Salmon provides fast and bias-aware quantification of transcript expression***
Patro et al (2017) Nature Methods (2017) doi:10.1038/nmeth.4197

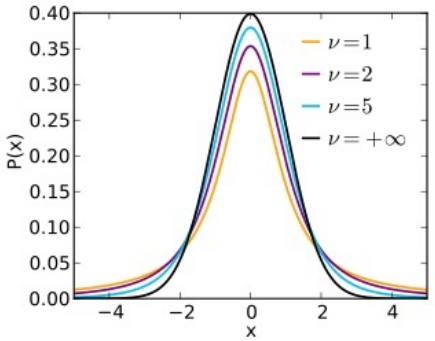
# T-test



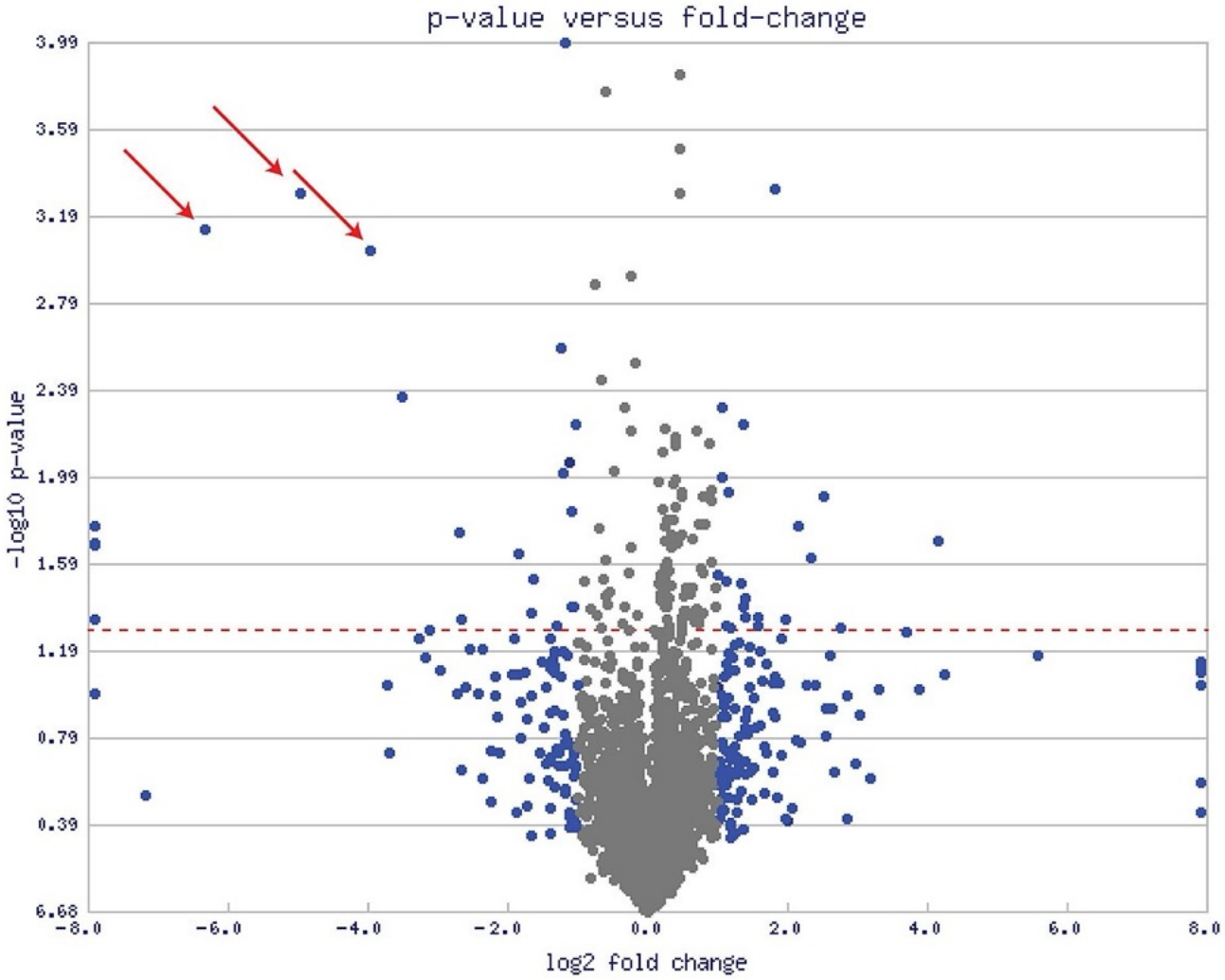

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Where:

- $\bar{X}_1$ and $\bar{X}_2$ are the means of the two samples.
- $s_1^2$ and $s_2^2$ are the variances of the two samples.
- $n_1$ and $n_2$ are the sizes of the two samples.



# Volcano Plot




**https://en.wikipedia.org/wiki/Volcano_plot_%28statistics%29**

# Annotation Summary

- Three major approaches to annotate a genome
  1. Alignment:
     - Does this sequence align to any other sequences of known function?
     - Great for projecting knowledge from one species to another
  2. Prediction:
     - Does this sequence statistically resemble other known sequences?
     - Potentially most flexible but dependent on good training data
  3. Experimental:
     - Lets test to see if it is transcribed/methylated/bound/etc
     - Strongest but expensive and context dependent



- Many great resources available
  - Learn to love the literature and the databases
  - Standard formats let you rapidly query and cross reference
  - Google is your number one resource ☺